

PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

CAPSTONE 2

**Automated Sales Monitoring and Resource Management System :
Optimizing Operations at Q-Solutions**

In partial fulfillment for the requirements in Capstone 2

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CHAPTER 1

INTRODUCTION

Introduction

The opening chapter of this study establishes the comprehensive intellectual, operational, and contextual foundation upon which the entirety of the research rests. It articulates the environmental conditions that make the proposed system both necessary and timely, defines with precision the specific operational problems the system is designed to solve, states the objectives that guide its development, delineates the boundaries of its functional scope, establishes the significance of its contributions to multiple categories of stakeholders, defines the key technical and operational terms employed throughout the study, and presents the conceptual framework that organizes the logical relationship between the system's inputs, processes, and outputs.

Each section of this chapter has been developed with the specific operational context of Q-Solutions at its center, ensuring that every element of the foundational framework is directly connected to the real-world problems and opportunities that motivate this research. The chapter does not treat these foundational elements as mere academic formalities to be dispatched quickly before arriving at the substantive technical content of later chapters. Rather, it develops each element with the depth and specificity that the complexity of the operational problem genuinely demands, recognizing that a rigorous, well-grounded introduction is the prerequisite for all analytical, technical, and empirical work that follows.

The chapter proceeds through seven major sections. Section 1.1 develops the background of the study, situating the Q-Solutions operational context within the broader landscape of digital access inequality, the evolution of computer rental establishments in the Philippine context, the specific operational realities of small digital service businesses, and the inadequacy of the manual administrative frameworks that currently govern Q-Solutions' operations. Section 1.2 states the specific operational problems that motivate the research, articulating each problem with sufficient precision to establish the criteria against which the proposed system's

effectiveness can be evaluated. Section 1.3 states the general and specific objectives that govern the system's design and development. Section 1.4 delineates the scope and limitations of the study, defining clearly what the system does and does not encompass. Section 1.5 articulates the significance of the study for each of the primary stakeholder communities affected by its outcomes. Section 1.6 provides formal definitions for the key technical and operational terms employed throughout the study. Section 1.7 presents the conceptual framework — organized around the Input-Process-Output model — that governs the study's analytical and methodological structure.

Together, these seven sections constitute a complete and rigorous intellectual foundation for the research that follows, establishing not merely what the system is and what it does, but why it matters, to whom it matters, and what intellectual framework organizes the understanding of its development and impact.

1.1 Background of the Study

1.1.1 The Digital Divide and the Structural Importance of Access Infrastructure

In the contemporary global environment, the capacity to access and effectively utilize digital technology has become one of the most consequential determinants of individual opportunity, economic mobility, and social participation. The digital revolution that has unfolded over the past three decades has fundamentally transformed the landscapes of education, commerce, governance, healthcare, and interpersonal communication, creating systems and processes that increasingly presuppose digital access as a baseline condition for meaningful participation in modern society.

Academic institutions at every level — from primary and secondary schools to universities and professional training programs — have progressively integrated digital platforms into their core educational delivery systems. Curricula are delivered through learning management systems that require reliable internet access to navigate. Assignments are submitted through online portals that require functional computing devices to complete. Research requires the ability

to search and access scholarly databases, download and annotate documents, and synthesize information from multiple simultaneously open browser sessions. Communication with faculty and peers occurs increasingly through email, messaging platforms, and video conferencing tools that demand stable connectivity and capable hardware. Assessment increasingly takes place through online testing platforms that impose specific browser and hardware requirements on examinees. For students at all levels of the Philippine educational system, the inability to reliably access capable computing resources is not merely an inconvenience — it is a fundamental barrier to educational participation and achievement that directly affects academic performance, degree completion, and ultimately, long-term professional opportunity.

The employment landscape presents analogous challenges. Employers across virtually every sector have digitized their recruitment and selection processes, posting vacancies on online job boards, receiving applications through web-based applicant tracking systems, and conducting preliminary screening through digital assessments and video interviews. Job seekers who lack reliable computing access cannot effectively navigate these digitized hiring processes, cannot create and format professional application documents, and cannot maintain the digital professional presence — through platforms like LinkedIn and professional email — that contemporary employers increasingly expect. The digital requirements of employment do not end with recruitment; once employed, workers across a broad range of industries and occupational levels are expected to use productivity software, communicate through digital platforms, and generate digital work products as routine elements of their job performance.

Governance and civic participation have followed the same digitization trajectory. National and local government agencies in the Philippines have progressively migrated public services to online delivery channels, requiring citizens to file tax returns through the Bureau of Internal Revenue's electronic filing systems, to apply for government identification documents through online portals, to access social welfare program information and applications through digital platforms, and to communicate with regulatory bodies through email and web-based service request systems. Citizens who lack reliable digital access are increasingly disadvantaged in their ability to exercise civic rights and access public services to which they are entitled, creating a dimension of digital exclusion that extends beyond economic disadvantage into the domain of democratic participation and civic equity.

Against this backdrop of pervasive and deepening digitization, the persistence of a significant and consequential digital divide — the gap between those who have reliable, capable access to digital technology and those who do not — represents one of the most pressing equity challenges of the contemporary era. This divide is not, critically, merely a question of whether an individual owns a computing device of any kind. The contemporary digital divide must be understood across multiple dimensions that together determine the quality and productivity of an individual's digital access: the capability of available devices to support the full range of productivity applications that contemporary educational, professional, and administrative tasks demand; the reliability and speed of internet connectivity sufficient to support bandwidth-intensive applications including video communication, cloud-based software, and large file transfer; the stability of access sufficient to allow extended productive sessions without interruption; and the ergonomic and peripheral suitability of the access environment for the sustained, focused work that productive computing requires.

A smartphone, however capable by contemporary consumer mobile device standards, cannot serve as an adequate substitute for a dedicated workstation for the range of tasks that constitute productive digital work. The physical constraints of a smartphone's screen size, touch interface, processing capacity, and memory make it fundamentally unsuitable for extended document creation and editing, complex spreadsheet construction and analysis, professional graphic design and image editing, programming and software development, academic research requiring simultaneous management of multiple documents and reference sources, or administrative work involving database interaction and financial record keeping. These tasks — which constitute the core of the productive digital work that students completing assignments, job seekers preparing applications, entrepreneurs managing business operations, and professionals fulfilling work requirements need to accomplish — require the processing power, memory capacity, screen size, physical keyboard and mouse interfaces, and peripheral connectivity that only a dedicated desktop or laptop computer can reliably provide.

For the tens of millions of Filipino households for whom the purchase and maintenance of a personal desktop or laptop computer represents an economically prohibitive investment — one that would consume a disproportionate share of a monthly income already fully committed to meeting basic necessities — computer rental centers, internet cafés, and digital service hubs serve as the primary and often exclusive means of accessing the productive computing capabilities that their educational, professional, and administrative needs demand. The economic arithmetic of this situation is straightforward but powerful: a household whose monthly income cannot sustain the capital expenditure of a computer purchase and the ongoing costs of home internet subscription can nevertheless access capable computing and connectivity resources on a pay-per-use basis at a computer rental establishment, paying only for the specific time they need rather than bearing the full cost of permanent ownership. This pay-per-use model transforms what would otherwise be a prohibitively expensive capital investment into an accessible, metered service that even economically constrained users can afford for the durations necessary to complete their specific tasks.

The social significance of this accessibility function cannot be overstated. Computer rental establishments occupy a position in the technology access ecosystem that is simultaneously commercially modest — small businesses operating on thin margins in competitive local markets — and socially critical — serving as the primary gateway to full digital participation for a substantial portion of the communities in which they operate. The quality of their operations — the reliability of their hardware, the speed of their internet connectivity, the breadth of their service offerings, and the efficiency and honesty of their administrative management — has direct consequences for the quality of digital access available to a large segment of the Philippine population that lacks alternatives. A poorly managed establishment with unreliable equipment, slow connectivity, inaccurate billing, and disorganized operations does not merely fail commercially; it fails the community it serves by delivering inadequate access to a resource on which community members depend for their educational and professional advancement.

1.1.2 The Philippine Context: Digital Access Patterns and the Role of Computer Rental Establishments

The Philippines presents a particularly instructive case for studying the role of computer rental establishments in the digital access ecosystem, because the country's digital landscape combines high mobile internet penetration with persistently low rates of personal computer ownership and fixed broadband subscription — creating conditions under which a large proportion of the digitally active population depends on shared access facilities for their productive computing needs. The Philippines has consistently ranked among the world's most active social media and mobile internet user populations, with smartphone penetration driven by the relative affordability of entry-level and mid-range Android devices and the availability of prepaid

These metrics paint a picture of a highly digitally engaged society — one in which citizens are comfortable with digital platforms and actively use them for communication, entertainment, and information consumption. However, this mobile-centric digital engagement pattern masks a significant gap between casual consumer digital use — social media, messaging, entertainment streaming — and the more demanding productive digital use that educational and professional advancement requires.

The personal computer ownership rate in the Philippines, while growing, remains substantially lower than in higher-income countries and lower than the smartphone penetration rate would suggest if these devices were adequate substitutes for productive computing. Fixed broadband subscription rates similarly lag behind mobile internet penetration, reflecting both the infrastructure investment required to extend fixed broadband networks to lower-density residential areas and the subscription cost barriers that make fixed broadband unaffordable for lower-income households even where the infrastructure exists. The result is a digital access profile characterized by broad but shallow mobile connectivity — widely available but limited in its capacity to support the full range of productive digital tasks that contemporary life demands. This access profile creates a structural market for computer rental establishments and digital service hubs across the country, particularly in the peri-urban and provincial areas where personal computer ownership rates are lowest and the gap between mobile access capability and productive computing need is widest. In these communities, computer rental establishments are

not marginal or declining institutions serving a niche market of individuals who cannot afford personal devices. They are central community infrastructure — analogous in their social function to public libraries, serving as the primary point of access to the information, tools, and services that community members need for their advancement.

Q-Solutions operates in precisely this context. Its customer base reflects the full diversity of the productive digital access needs that characterize a Philippine community with moderate personal computer ownership rates: students completing academic assignments and conducting research, job seekers preparing applications and accessing job boards, small business owners managing financial records and communicating with suppliers and customers, government service recipients filing documents and accessing online portals, freelancers completing work for remote clients, and community members fulfilling a wide range of personal and professional digital tasks that require more than a smartphone can deliver. This diversity of customer needs is reflected in the breadth of

Q-Solutions' service portfolio — which extends beyond basic PC rental and internet access to encompass document printing, photocopying, and other digital services — and creates the operational complexity that makes automated management both challenging and necessary.

1.1.3 The Evolution of Computer Rental Establishments: From Internet Café to Digital Service Hub

The history of internet cafés and computer rental establishments in the Philippines traces a trajectory of significant transformation over more than two decades — from the earliest days of public internet access in the late 1990s through the dramatic sector expansion of the first decade of the twenty-first century to the more complex, multi-service digital hub model that characterizes the most viable establishments operating in the contemporary market environment.

In the earliest phase of the sector's development, roughly spanning the period from 1997 to 2005, internet cafés competed primarily on internet access speed and hardware recency — metrics that were relatively straightforward for customers to evaluate and that created clear competitive differentiation in markets where both internet speeds and hardware quality varied dramatically across providers.

The customer base of this period consisted largely of early technology adopters, gaming enthusiasts, and professionals who used internet access for email and web browsing — a relatively limited set of use cases that did not demand the operational complexity of a modern multi-service establishment. Management of these early establishments was primarily a matter of counting occupied workstations, tracking session durations with basic timing tools, and collecting the appropriate hourly fee at the end of each session.

The second phase of the sector's evolution, roughly spanning the period from 2005 to 2015, was characterized by rapid expansion driven by the democratization of internet access, the rise of online gaming as a mass-market entertainment category, and the growing integration of digital tools into educational curricula and business operations. The number of computer rental establishments expanded dramatically during this period, driving competition that progressively eroded the simple time-and-access differentiation of the early period. As internet connectivity became more widely available at more standardized speeds, and as the baseline quality of consumer hardware improved and converged toward acceptable standards across the industry, competitive differentiation shifted toward service breadth and operational quality.

Establishments that offered only basic internet access faced increasing pressure from competitors that supplemented computing and connectivity with document printing, photocopying, scanning, and other digital service capabilities. The commercial logic of service expansion was straightforward: a customer who needed both internet access and document printing services would prefer an establishment that offered both, because it allowed them to meet multiple needs in a single visit rather than traveling to separate locations. For the establishment, the addition of each service category created a new revenue stream that distributed fixed costs across a broader revenue base and increased the average transaction value per customer visit.

This competitive dynamic drove the progressive evolution of the most resilient establishments from single-service internet cafés into multi-service digital service hubs — establishments whose value proposition rested not on the superiority of any single service but on the comprehensive convenience of a one-stop digital service experience. This evolution is commercially rational and customer-beneficial, but it introduces operational complexity that single-service establishments do not face.

A multi-service digital hub must simultaneously manage multiple distinct service categories, each with its own pricing structure, resource consumption pattern, and administrative requirements. PC rental sessions must be timed and billed on a duration basis.

Print jobs must be logged, priced by specification, and reflected in consumable inventory. Photocopying must be tracked and billed by unit. The administrative framework that must manage these multiple concurrent service streams — tracking which customers have consumed which services at what quantities, computing the total charges for multi-service transactions, and maintaining complete and accurate records of all transactions — is vastly more complex than the single-dimension time-and-rate billing of a basic internet café. And yet, many establishments at this stage of evolution continue to rely on the same essentially manual administrative frameworks that served adequately when operations were simpler — extending their paper logs, calculator billing, and manual record keeping into increasingly complex operational environments for which those tools were never designed.

The third and current phase of the sector's evolution is characterized by dual pressures: the continuing commoditization of basic computing and connectivity access as smartphones and mobile data become more capable and affordable, and the increasing operational complexity of the multi-service model as customer expectations rise and service portfolios expand. Establishments that fail to respond to either of these pressures — either by differentiating their value proposition beyond basic access provision or by professionalizing their operational management to deliver consistent service quality efficiently — face declining commercial viability.

Q-Solutions represents an establishment at the inflection point of this phase: operationally complex enough that manual management has become a serious strategic liability, and commercially positioned to benefit significantly from the operational transformation that effective automation can deliver.

1.1.4 The Specific Operational Context of Q-Solutions

Q-Solutions is a multi-service digital service hub that offers customers access to personal computer workstations with internet connectivity and standard productivity software, document printing services across multiple paper sizes and ink type configurations, photocopying services, and potentially additional digital service offerings that address the diverse needs of its community customer base. The establishment operates in a commercial environment that requires it to simultaneously deliver reliable, quality service to customers and manage the operational complexity of its multi-service portfolio efficiently and accurately.

The workstation inventory of Q-Solutions consists of a defined number of desktop personal computers equipped with current productivity software suites, internet browsers, and the hardware peripherals — keyboards, mice, and headsets — that customers require for their rental sessions.

These workstations represent the primary revenue-generating assets of the business, and their effective utilization — maximizing the proportion of available session hours that are occupied by paying customers while maintaining the hardware in reliable working condition — is the central operational challenge of PC rental management.

The printing and photocopying services offered by Q-Solutions generate revenue on a per-job and per-copy basis, creating a supplementary revenue stream that both adds value for customers and diversifies the business's income sources. These services require the management of consumables — paper stock across multiple size formats, printer ink or toner in multiple colors — whose consumption must be tracked both for billing purposes and for inventory management purposes. The financial contribution of these services, while individually modest on a per-transaction basis, is collectively significant and represents revenue that is particularly vulnerable to the underrecording and billing errors that characterize manual transaction management.

The customer base of Q-Solutions reflects the community it serves — primarily students, job seekers, small business operators, and working adults who need capable computing and digital service access for educational, professional, and personal purposes. This customer profile creates

predictable demand patterns: pronounced peaks during after-school and after-work hours, moderate demand during midday hours, and reduced demand during early morning and late evening periods. Managing staff deployment, workstation readiness, and consumable availability in alignment with these demand patterns requires operational data — workstation utilization rates, peak hour transaction volumes, service mix distributions — that the current manual administrative system cannot reliably generate.

The staff complement of Q-Solutions consists of a lean team whose individual members must perform multiple operational functions simultaneously — managing customer interactions, monitoring active sessions, processing print and copy jobs, handling checkout transactions, and maintaining the physical environment of the establishment. This operational model, common in small service businesses, creates a context in which administrative efficiency has a direct and significant impact on service quality: every minute that staff devote to the manual administrative tasks of session logging, billing calculation, and record keeping is a minute unavailable for the customer-facing activities that determine the customer experience.

1.1.5 The Inadequacy of Manual Administrative Frameworks

Given the operational conditions described above, the inadequacy of manual administrative frameworks for managing a multi-service digital hub like Q-Solutions is not a matter of minor inconvenience but of fundamental operational incapacity. The standard procedures that Q-Solutions has historically employed — handwritten session logs, standalone timing devices, manual billing calculations, and paper-based financial records — may have been adequate when the business was simpler and smaller, when transaction volumes were lower, and when the operational demands on staff were less intense. As the business has matured and its operational complexity has grown, however, these frameworks have become structurally inadequate — unable to deliver the accuracy, consistency, security, and analytical capability that the business needs to operate effectively in the contemporary competitive environment.

The handwritten session log is the foundational tool of the manual session management framework. In this approach, a staff member records the start time of each customer's session by hand in a physical logbook when the customer occupies a workstation, monitoring elapsed time through periodic checks of a wall clock or personal device, and records the end time when the

customer concludes their session. The inherent limitations of this approach begin with its dependence on the moment-to-moment attention of a human operator who is simultaneously managing multiple other tasks. In a business environment where a single staff member may be managing six, eight, or ten concurrent sessions during peak hours, the probability that every session will have its start time accurately recorded at the precise moment it begins is low. The probability that every session's expiration will be noticed and acted upon promptly is similarly low. And the probability that accumulated errors across multiple sessions will go undetected until the end of the business day — at which point correction is no longer practically possible — is high.

The manual billing calculation process compounds these timing inaccuracies with arithmetic errors and pricing inconsistencies. Computing the correct total for a multi-service transaction requires applying multiple different pricing rules to multiple different consumption quantities and summing the results correctly — a process that demands careful attention and is vulnerable to errors of arithmetic, memory, and interpretation under any conditions. During peak business hours, when staff members are processing multiple checkouts in rapid succession while simultaneously managing active sessions and responding to customer inquiries, the conditions for careful, accurate manual calculation are at their worst — precisely when the volume of calculations being performed is at its highest.

The paper-based record system adds a further layer of inadequacy in its fundamental vulnerability as a data storage medium. Physical records can be damaged or destroyed by water, fire, or physical deterioration. They offer no protection against unauthorized access or modification — any person with physical access to the logbook can alter its contents, with no mechanism for detecting or proving the alteration. They cannot support the data retrieval and aggregation operations that generate analytical insight — finding a specific historical record requires manual searching, and computing any aggregate metric requires manual tallying of individual entries. And they create a reporting process that is inherently delayed and labor-intensive, requiring manual compilation of data from multiple records to produce even a basic daily financial summary.

These inadequacies, individually significant, are in combination structurally disabling for a business that aspires to operate with the financial integrity, operational efficiency, and management capability that its commercial context demands. They represent not merely inefficiencies to be optimized but fundamental failures to be replaced — and it is this replacement that the proposed system is designed to achieve.

1.1.6 The Case for Automation: Evidence from the Research Literature

The operational problems documented above at Q-Solutions are not unique to this establishment. They are characteristic of a broad category of small digital service businesses that share similar operational profiles — multi-service offerings, peak-hour demand concentration, lean staffing, thin margins, and manual administrative frameworks — and the research literature on these businesses documents the same patterns of operational failure and the same potential for improvement through automation that motivate the present study.

Research on administrative efficiency in computer rental and internet café operations consistently identifies manual time tracking as the most significant single source of operational error and revenue leakage. Williams and Thompson (2021) documented billing discrepancy rates of fifteen percent in manually managed computer rental establishments, attributing the majority of these discrepancies to timing errors arising from inaccurate manual session duration recording.

Santos and Reyes (2020) found that automated session management systems reduced checkout processing time by up to sixty percent compared to manual methods, with corresponding improvements in customer throughput and staff operational capacity during peak periods. Villanueva and Mercado (2022) specifically documented the trust and satisfaction implications of billing transparency in Philippine internet café operations, finding that customers who received itemized, system-generated billing records expressed significantly higher confidence in the fairness of their charges than customers who received manually computed bills.

The research literature on database-driven management systems for small service businesses similarly documents the transformative potential of structured digital data management. Tan and Reyes (2019) found that small businesses that transitioned from paper-based to digital record

systems reported significant improvements across multiple dimensions of financial management — reduced revenue leakage, faster report generation, improved audit capability, and enhanced ability to identify and respond to operational trends. Fernandez and Cruz (2021) documented the security improvements associated with database-driven record management, finding that the audit logging and access control capabilities of digital systems substantially reduced both the incidence and the impact of record manipulation compared to paper-based systems.

The convergence of these research findings with the specific operational realities documented at Q-Solutions provides strong empirical justification for the development and implementation of the proposed automated management system. The research literature does not merely suggest that automation could improve Q-Solutions' operations — it documents, with quantified precision, the specific improvements that systematic automation of session management, billing computation, and data management can deliver in operational contexts closely analogous to Q-Solutions'.

1.2 Statement of the Problem

1.2.1 The General Problem

The operational landscape of Q-Solutions, as described comprehensively in Section 1.1, is characterized by a set of deeply interconnected administrative failures that collectively undermine the business's financial integrity, operational efficiency, service quality, and competitive viability. These failures are not isolated incidents or occasional disruptions attributable to specific staff errors or unusual operational circumstances. They are systemic properties of the manual administrative framework that currently governs the business's operations — structural features of a management approach that is fundamentally misaligned with the operational complexity of a multi-service digital hub operating under contemporary market conditions.

They recur with regularity and predictability across every shift, every business day, and every financial reporting cycle, because they are consequences not of individual failure but of systemic inadequacy. The interconnected nature of these failures is a critical feature of the problem that any proposed solution must address comprehensively. Manual session timing inaccuracies directly cause billing computation errors, because incorrect session duration data is the input to the billing calculation. Billing computation errors directly cause revenue leakage and customer disputes, because incorrectly computed bills either undercharge customers (causing direct revenue loss) or overcharge them (causing dissatisfaction and conflict). Paper-based record keeping directly causes reporting inadequacy, because the analog storage medium cannot support the data aggregation and retrieval operations that systematic financial reporting requires. And peak-hour operational bottlenecking directly exacerbates every other failure category, because the conditions of cognitive overload and time pressure that peak-hour manual management creates amplify the frequency and severity of errors in every administrative process. A solution that addresses only one or two of these failure categories — improving billing calculation, for example, without addressing session timing accuracy or data management — will deliver partial improvement at best.

Only a comprehensive solution that addresses the complete operational system — replacing manual session tracking, billing computation, record keeping, and reporting with automated equivalents simultaneously — can break the interconnected cycle of failures and deliver the operational transformation that Q-Solutions requires.

The general problem that this study addresses may therefore be stated as follows:

How can the development and implementation of the Automated Sales Monitoring and Resource Management System optimize the daily operations of Q-Solutions by eliminating the inefficiencies, financial inaccuracies, and data vulnerabilities associated with its traditional manual administrative workflows, and by providing management with the real-time operational visibility and analytical capabilities necessary for effective business decision-making?

This general problem statement captures the full scope of the study's ambition — not merely to replace one or two manual processes with digital equivalents, but to comprehensively transform the administrative infrastructure of the business in ways that address the entire constellation of

operational problems that manual management has created. The specific problems enumerated in the following subsections represent the discrete dimensions of this general challenge, each demanding targeted solutions that together constitute the comprehensive operational transformation the proposed system delivers.

Problem 1: Inefficient and Inaccurate Real-Time Session Time Tracking

The most immediate and visible operational failure at Q-Solutions is the chronic inadequacy of its current session time tracking method. In the present manual system, the complete sequence of session management — from initiation through monitoring to expiration and checkout — depends entirely on the vigilance, memory, and availability of a human staff member who is simultaneously managing multiple other operational responsibilities.

The specific failure modes of this approach are numerous and well-documented in the operational context. Session initiation recording fails when the staff member is occupied with another task at the precise moment a customer occupies a workstation — a common occurrence during peak hours when multiple customers may be arriving, completing sessions, and requesting services simultaneously. The resulting start time recorded in the logbook reflects not the actual session start but the moment the staff member was able to attend to the recording task, introducing a systematic underestimation of session durations that translates directly into underbilling. Session duration monitoring fails because the sustained parallel attention required to track the remaining time on multiple concurrent sessions simultaneously — each started at a different moment and having a different purchased duration — exceeds the cognitive capacity of any individual operating under peak-hour conditions.

Session expiration goes undetected when the staff member's attention is directed elsewhere, allowing sessions to run beyond their purchased duration without the customer being prompted to pay for the extension — a direct and recurring source of revenue leakage. And session conclusion recording inherits the same accuracy limitations as initiation recording, particularly when the rush of a busy period pushes record-keeping tasks behind real-time events.

The cumulative financial impact of these tracking failures is significant. Every session that runs five minutes beyond its purchased duration without an extension charge being collected represents a loss of approximately one-twelfth of the hourly rate. Multiplied across the number of sessions processed in a typical business day and compounded across the operating year, these individually small losses aggregate into a substantial annual revenue leakage figure. The research literature cited above — particularly Williams and Thompson's (2021) documented fifteen percent billing discrepancy rate in comparable establishments — suggests that the magnitude of this leakage is likely to represent a meaningful proportion of the business's net operating margin.

Beyond the financial dimension, timing inaccuracies create a customer service problem that is qualitatively as significant as its financial dimension. Customers who believe they have been charged for time beyond their actual session duration — whether or not that belief is correct — experience a breakdown of trust in the establishment that manual systems provide no reliable mechanism to resolve.

Without an objective, tamper-evident record of session timing, disputes about session duration cannot be conclusively resolved, and the staff member's good-faith account of the manually recorded times cannot substitute for the objective evidence that an automated system provides.

How can workstation session durations be monitored automatically and in real time, with precision sufficient to eliminate manual timing discrepancies, prevent unauthorized session extensions, and provide a complete, accurate, tamper-evident session log for billing and audit purposes?

Problem 2: Inconsistent and Error-Prone Billing Computations

The billing computation process at Q-Solutions involves the systematic application of multiple pricing variables to multiple service consumption quantities to produce a single transaction total that accurately reflects everything a customer has consumed during their visit. For a customer who has used a PC rental workstation and also requested document printing and photocopying services — a typical multi-service transaction — the accurate computation of the total charge requires: retrieving the precise session duration from the session log, applying the applicable per-hour rental rate (potentially prorated for partial hours at the business's configured rounding

rule), determining the specifications of each print job (paper size and ink type), applying the applicable per-page rate for each job specification, counting the number of photocopies, applying the applicable per-copy rate, summing all component charges, and applying any configured discounts or promotions. This multi-step, multi-variable computation must be performed correctly for every checkout transaction processed, under whatever operational conditions exist at that moment. The structural problem with manual billing computation is not that any individual computation is impossibly difficult. A single transaction, computed carefully at a quiet desk with no competing demands, is within the capacity of any numerically literate staff member to calculate accurately. The structural problem is that the conditions of careful, uninterrupted attention that accurate manual calculation requires are precisely the conditions that peak-hour operations at a busy digital service hub systematically prevent. During peak hours, when transaction volumes are highest and the pressure on staff performance is most intense, the cognitive load of simultaneous session monitoring, customer service, and operational management leaves insufficient attentional resources for the careful computation that accurate billing demands. The result is a predictable and recurring pattern of arithmetic errors, pricing misapplications, and computational shortcuts that produce billing inaccuracies across a significant proportion of peak-hour transactions.

The inconsistency introduced by staff variability compounds the frequency problem. Different staff members may apply slightly different interpretations of partial-hour rounding rules. The same staff member may apply the rules differently on different days or at different points in their shift.

Without a formal computational record attached to each transaction, the basis on which any specific bill was computed cannot be reconstructed after the fact, making disputes about billing accuracy impossible to resolve definitively and making systematic quality control of billing accuracy impossible to implement. The aggregate financial impact of billing computation errors operates in both directions — errors that undercharge customers directly reduce revenue, while errors that overcharge customers create disputes and reduce customer satisfaction — but the direction of bias in manually computed bills under conditions of cognitive overload tends systematically toward undercharging, as staff members under time pressure tend to err on the

side of the lower estimate when uncertain about a specific computation, or omit service charges that they intended to add but forgot under the pressure of a busy period.

How can the system automatically compute transaction totals for all service types — including PC rental, document printing, photocopying, and any ancillary services — based on precisely defined pricing parameters, with perfect computational consistency across all transactions regardless of volume, time pressure, or operational complexity, and generate itemized billing records that provide a fully transparent and auditable basis for each charge?

Problem 3: Vulnerable and Inaccessible Data Storage

The paper-based record system that constitutes Q-Solutions' current primary data repository represents a data management framework that is inadequate across every dimension relevant to responsible and effective business data management. The dimensions of inadequacy are distinct but mutually reinforcing: physical vulnerability, security insufficiency, retrieval inefficiency, and analytical incapacity together constitute a data management profile that fails to meet even the most basic standards that effective business management requires. Physical vulnerability is the most viscerally obvious inadequacy. Paper logbooks are susceptible to damage or destruction through mechanisms that are common, varied, and essentially unavoidable in the ordinary operating environment of a commercial establishment. Water exposure — from spills, flooding, or humidity — can render handwritten records illegible. Fire, while less frequent, can destroy records irreversibly. Physical wear, particularly in the high-frequency-use environment of a busy business, degrades the legibility of records over time. There is no practical mechanism for creating backup copies of handwritten logbooks in real time, meaning that the loss of a physical record through any of these mechanisms is a permanent and irrecoverable loss of all the transactional data it contains.

Security is the second critical dimension of inadequacy. Physical paper records provide no meaningful protection against unauthorized access or alteration. The logbook, as a physical object, can be read by anyone who has access to it.

Entries can be modified, added, or removed by anyone with physical access and the motivation to do so. In a business environment where the accuracy and completeness of transaction records are the foundation of financial integrity, this absence of access control and modification protection creates serious vulnerability — both to accidental alteration (a spilled drink, a misfiled page) and to deliberate manipulation by anyone who might benefit from concealing a transaction or altering a billing record. The absence of any audit trail for record modifications means that alterations, once made, are typically undetectable and therefore also undeniable.

Accessibility and retrieval efficiency constitute a third category of inadequacy that has significant practical consequences for daily operations. Locating a specific historical transaction in a paper logbook requires physical page-by-page searching — a time-consuming manual process that interrupts other operational activities and becomes progressively more difficult as the volume of records grows. There is no mechanism equivalent to a database query for filtering records by date, workstation, staff member, or transaction amount. Every analytical question — what were yesterday's total revenues? which workstation generated the most sessions last week? how many print jobs were processed in the past month? — requires the manual tallying of individual records, a process that is both time-consuming and introduces its own layer of computational error.

Analytical incapacity is the fourth and strategically most consequential dimension of inadequacy. A paper-based record system cannot support the systematic data analysis that generates the operational insights necessary for evidence-based management. Management cannot use paper records to identify usage patterns that would optimize workstation deployment, to detect pricing optimization opportunities, to compare performance across periods, to identify potential irregularities in billing or revenue collection, or to project future resource requirements. The business operates in a state of analytical darkness — knowing its current operational state only approximately, understanding its historical performance only in broad outline, and making strategic decisions on the basis of intuition and memory rather than systematic data analysis.

How can transaction histories, session logs, customer billing records, and resource utilization data be stored in a centralized, secure, and structured digital repository that protects against unauthorized access and modification, enables rapid retrieval of specific records, and supports

the systematic aggregation and analysis of operational data for management reporting and decision-support purposes?

Problem 4: Administrative Bottlenecks During Peak Business Hours

The operational bottlenecking that occurs at Q-Solutions during peak business hours is the most immediately visible and experientially impactful manifestation of the inadequacy of manual administrative management. Peak periods — typically concentrated in the after-school and after-work time windows when customer demand is highest — create conditions of concurrent administrative demand that fundamentally exceed the capacity of any individual staff member to manage all required tasks accurately, promptly, and simultaneously.

The anatomy of a peak-hour administrative bottleneck at Q-Solutions unfolds predictably. As the peak period begins, customers arrive in clusters rather than at regular intervals — a behavioral pattern driven by the synchronization of school dismissal, office closing, and other temporal anchors in the community's daily schedule. Each arriving customer requires workstation assignment, session initiation logging, and potentially a brief operational orientation. Simultaneously, customers who arrived earlier in the afternoon are reaching the end of their purchased session durations — requiring the staff member to notice this, engage the customers about extension options, process extension requests, update session records, and handle payment for extensions. Print job requests arrive from customers at active workstations. Customers completing their sessions require checkout processing — session duration calculation, service charge tallying, billing computation, payment collection, and change processing.

And the operational environment of the establishment must be maintained — hardware issues noticed and addressed, equipment malfunctions handled, and the general service environment kept orderly. This simultaneous multitasking demand creates a condition that systems researchers refer to as cognitive overload — a state in which the total demand on an individual's attention and decision-making capacity exceeds the capacity of the individual to process all inputs and execute all required tasks accurately and promptly. Under cognitive overload, performance quality degrades in predictable ways: task switching becomes slower, error rates increase, less urgent tasks are deferred or neglected, and the individual's ability to maintain awareness of the

full operational state — all active sessions and their remaining durations, all queued service requests, all approaching session expirations — diminishes.

The cascading consequences of peak-hour administrative bottlenecks extend well beyond the staff member to affect every customer in the establishment. Customers waiting for checkout because a staff member is occupied with manual billing calculations experience service delays that directly degrade their satisfaction. Customers whose session expirations go unnoticed experience either the awkward discovery that their session has run beyond its purchased time or, if the staff member fails to notice, the implicit subsidy of unrecorded session time — which benefits them individually but represents a revenue loss for the business.

Customers seeking print services who cannot get prompt attention because staff capacity is consumed by administrative tasks experience service quality failures. And customers waiting to access a workstation who must wait longer than necessary because the checkout process for the previous occupant is slow experience a direct reduction in their own productive session time.

How can the system eliminate the administrative bottlenecks created by manual processes during peak business hours, reducing checkout processing time, enabling staff to manage higher transaction volumes without proportional increases in error rates or service quality degradation, and freeing staff capacity for the customer service functions that most directly determine customer satisfaction and operational quality?

Problem 5: Delayed and Inaccurate Financial Reporting

Effective management of any business — regardless of its size or industry — requires timely and accurate financial information. The ability to know, at any point during the operating day, what the current revenue total is; to compare current performance against historical baselines; to identify discrepancies between expected and actual revenue as they occur rather than after the fact; and to generate the periodic financial reports necessary for tax compliance, operational planning, and investor or lessor communication — all of these management functions depend on the availability of accurate, current financial data.

In the absence of this data, business management is reduced to retrospective assessment of past performance rather than informed, proactive management of current operations. For the management of Q-Solutions, the current manual system renders timely, accurate financial management essentially impossible in practice. Financial report generation requires the manual compilation of data from paper logbooks — an activity that cannot practically be performed during business hours without diverting staff from their operational responsibilities, must therefore be deferred to the end of the business day at the earliest, and produces results that are only as accurate as the underlying manual records from which they are compiled. Given the error rates documented in earlier sections — timing inaccuracies, billing computation errors, and missing or incomplete records — the financial summaries produced by manual end-of-day compilation are systematically inaccurate, reflecting not the actual operational performance of the business but the imperfect record of that performance captured in the manual system.

The delay between transaction occurrence and financial report availability is not merely an inconvenience — it has direct financial consequences. Revenue discrepancies that would be detectable and potentially correctable if identified in real time may become permanently unrecoverable by the time the end-of-day compilation identifies them.

A session that ran beyond its purchased duration without an extension being charged will not be identified until the end-of-day review — at which point the customer has long departed and the opportunity to collect the extension charge has passed. A billing error that understated a transaction total will not be identified until the daily reconciliation — at which point correction requires contacting the customer (if their contact information was recorded) or absorbing the loss.

Beyond the daily operations level, the inadequacy of the manual reporting system has strategic consequences. Without reliable period financial reports — weekly, monthly, and annual summaries of revenue, transaction volumes, and service category performance — management cannot make evidence-based decisions about pricing, staffing, capital investment, or service mix. They cannot identify whether a specific service category is growing or declining in revenue contribution. They cannot determine whether their current pricing structure is optimally calibrated to maximize revenue. They cannot project future resource requirements on the basis of

documented trends in demand. These strategic analytical capabilities — which a database-driven automated reporting system provides automatically and continuously — are precisely what the current manual system cannot deliver.

How can the system generate instantaneous, accurate financial summaries and operational analytics directly from comprehensive, real-time transactional records, enabling management to monitor business performance continuously throughout the operating day and make timely, evidence-based decisions about operations, staffing, resource allocation, and service management?

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study is to design, develop, and implement a fully functional Automated Sales Monitoring and Resource Management System for Q-Solutions, utilizing PHP for server-side application logic and MySQL for relational database management, with the overarching purpose of enhancing the operational efficiency, financial accuracy, data security, and management decision-support capability of the business across all dimensions of its daily operations.

This objective encompasses the complete lifecycle of the system — from the initial requirements analysis that translates operational needs into precise technical specifications, through the system design phase that transforms those specifications into an architectural blueprint, through the development phase that implements that blueprint in working software, through the testing phase that validates the software's correctness and performance, to the deployment and evaluation phases that confirm the system's effectiveness in the live operational environment.

It encompasses not only the technical work of system construction but the organizational work of operational integration — the training, workflow adaptation, and change management activities that must accompany technical implementation if the system's potential benefits are to be fully realized. The general objective is framed in terms of four specific dimensions of operational improvement — efficiency, accuracy, security, and decision-support capability — that

correspond directly to the four most consequential categories of inadequacy identified in the statement of the problem. This correspondence ensures that the general objective is not merely aspirationally broad but operationally specific: it commits the study to delivering improvements that are measurable against the documented operational failures that motivated the research, and against which the system's performance will be evaluated in the study's evaluation phase.

Objective 1: Develop an Intuitive Administrative Dashboard

The first specific objective is to design and implement a centralized, role-differentiated administrative dashboard that provides Q-Solutions staff and management with a unified, real-time operational view of all active sessions, available resources, and key financial indicators, accessible through standard web browsers on any networked device within the establishment's local area network.

The dashboard represents the primary interface between users and the system — the surface through which all operational interactions occur and through which all system-generated information is communicated to users. Its design quality therefore has a direct and significant impact on the overall effectiveness of the system: a dashboard that is confusing, cluttered, or slow to navigate undermines the operational efficiency improvements that the system is designed to deliver, regardless of how accurately the underlying computation and data management functions perform. Conversely, a well-designed dashboard that presents critical information clearly, enables the most common operational actions through minimal interaction steps, and organizes the full range of system functions in a logically structured and easily navigable hierarchy amplifies the effectiveness of every other system component by ensuring that the computational and data management work of the underlying modules is presented to users in a form they can immediately understand and act upon. The specific design requirements of the dashboard are derived from the operational context in which it will be used — a busy service environment where staff members are simultaneously managing multiple tasks, operating under time pressure during peak hours, and frequently interrupted by customer interactions and operational events. In this context, interface simplicity is not merely an aesthetic preference but an operational necessity. Every additional screen element that does not contribute directly to the user's immediate operational task increases the cognitive load of using the interface.

Every navigation step that could be eliminated reduces the time required to complete an operational action. Every ambiguous label or non-intuitive interaction pattern increases the likelihood of user error. The dashboard must therefore be designed to the standard of minimal necessary complexity — providing everything users need to perform their roles effectively and nothing unnecessary that would impede their performance. The role-differentiated architecture of the dashboard must implement a meaningful and operationally appropriate distinction between the Administrator and Staff user categories. The Staff interface must provide full, unimpeded access to all operational transaction-processing functions — session initiation, monitoring, extension, checkout processing, and service recording — while appropriately restricting access to configuration, historical record modification, and system administration functions that are outside the operational scope of the Staff role. The Administrator interface must provide all Staff-level capabilities plus the additional functions of system configuration management, user account administration, historical data review, and full reporting — organized in a manner that allows Administrators to move efficiently between operational oversight and administrative management activities.

Objective 2: Implement a Real-Time Automated Timer Module

The second specific objective is to engineer and implement an automated session timer module that tracks the duration of each PC rental session with precision to the second, maintains accurate session state across the full duration of each session from initiation through any extensions to final checkout, displays continuously updated session status information on the administrative dashboard for all active sessions simultaneously, and generates automatic alerts when sessions approach or reach their purchased duration limit.

This module is the operational cornerstone of the entire system — the source of the precise timing data upon which all billing accuracy ultimately depends — and its technical design must therefore prioritize reliability, accuracy, and resilience above all other considerations. A timer module that operates correctly under normal conditions but fails under high concurrency, or that loses state accuracy when the browser is refreshed, or that is vulnerable to client-side manipulation, does not deliver the billing accuracy that the system is designed to provide. The timer module must be designed to operate correctly under all anticipated operational conditions, including peak-hour concurrency loads, routine browser interactions, and the various hardware

and software events that may occur in the operational environment. The architectural cornerstone of the timer module's design is the principle of server-side state authority. The authoritative record of each session's duration and current status must be maintained in the database, not in the client browser or workstation. The client-side timer display — the visual countdown or count-up shown on the dashboard — must be treated as a presentation layer that is continuously synchronized with the server-side authoritative state, not as an independent timing mechanism whose state is authoritative. This design principle ensures that the timer continues to operate accurately regardless of client-side events — browser refreshes, tab switches, workstation reboots — that would disrupt a client-side timing mechanism. It also ensures that the session duration data used for billing computation is drawn from the same authoritative server-side record that has been reliably maintained throughout the session, rather than from a client-side calculation that may have diverged from the true elapsed time.

The extension processing capability of the timer module must handle session extensions seamlessly and accurately. When a customer wishes to extend their session, the staff member must be able to record the extension through a simple dashboard interaction that updates the session record in the database, adjusts the dashboard display to reflect the extended duration, and creates an extension event record that captures the timing and amount of the extension for billing and audit purposes.

The billing computation for a session that has been extended must accurately reflect the total duration — both the original purchased period and the extension — ensuring that the final checkout charge correctly accounts for the complete session length.

Objective 3: Automate Billing and Fee Computation

The third specific objective is to develop a billing computation engine that automatically calculates the total charges for every customer transaction — encompassing PC rental fees based on session duration, document printing charges based on job specifications, photocopying fees, and any applicable service charges — by applying the business's configured pricing parameters to the service consumption data generated by the session tracking and service logging modules. The billing engine must produce itemized transaction records that clearly present each service component's charges, the applicable rates, and the computed subtotals, providing transparency

that supports customer confidence and provides a detailed audit trail for financial management and dispute resolution.

The billing computation logic must handle the full complexity of Q-Solutions' pricing structure with complete accuracy. For PC rental billing, this includes correct proration of partial-hour sessions according to the business's configured rounding rules — whether partial hours are rounded to the nearest half-hour, billed in fifteen-minute increments, or handled according to any other configured rule. For document printing billing, this includes the correct application of per-page rates that vary by paper size and ink type — differentiating, for example, between letter-size black-and-white printing, A4 color printing, legal-size printing, and any other specification categories the business's service portfolio encompasses. For photocopying billing, this includes the correct application of per-copy rates with any applicable volume discounts. For any promotional or discount programs the business offers, this includes the correct identification and application of applicable discounts based on configured eligibility criteria.

The pricing parameter architecture of the billing engine must be fully configurable through the administrative interface, allowing management to update any pricing parameter — hourly rental rates, per-page printing rates, per-copy photocopying rates, discount structures — without requiring any modification to the application's source code. This configurability is essential to the system's long-term utility: a system whose pricing logic is hard-coded into the application and can only be updated by a developer is not practically maintainable by a small business that may not have reliable access to technical development resources. By externalizing all pricing parameters to configurable database records, the system ensures that management can maintain accurate billing logic as the business's pricing policies evolve, without technical dependency on a developer for routine parameter updates.

Objective 4: Establish a Secure Centralized Database

The fourth specific objective is to design and implement a normalized MySQL relational database that serves as the secure, persistent, and comprehensive data repository for all information generated by the system's operation — including session records, transaction histories, billing details, workstation inventory, user account information, system configuration parameters, and audit logs. The normalization of the database schema must follow established

third normal form design principles, eliminating data redundancy and ensuring that relationships between entities — sessions, transactions, workstations, service types, staff members, and billing records — are correctly and completely represented through foreign key relationships that enforce referential integrity across the complete dataset. The normalization discipline is particularly important for the financial data that constitutes the core of the system's data — transaction records and billing line items — where data redundancy would create the risk of update anomalies that could introduce inconsistencies between different instances of the same data element.

The security architecture of the database must implement multiple complementary layers of protection. Authentication controls must ensure that only authorized system accounts can establish database connections. Permission controls must ensure that each database account has only the access permissions necessary for its specific function — application accounts can perform the data manipulation operations required by the application, but cannot perform administrative operations such as schema modification or user account management. Audit logging must capture critical database events — record insertions, modifications, deletions, and access attempts by privileged accounts — providing an accountability trail that is essential for detecting and investigating potential irregularities in financial records.

The indexing strategy for the database must balance read performance — the speed of the queries that generate dashboard displays and reports — with write performance — the speed of the insert and update operations that record new transactions in real time. For a system that must respond to user interactions within sub-second time frames, query performance is a critical operational property, and the indexing design must be carefully planned to ensure that the most frequently executed queries — session status retrievals, dashboard data updates, and report queries — can be executed efficiently even as the database grows to contain the complete transaction history of the business's operations.

Objective 5: Integrate an Automated Analytics and Reporting Engine

The fifth specific objective is to design and implement a reporting module that transforms the raw transactional data stored in the database into structured, readable financial and operational

reports that provide Q-Solutions management with the decision-support information they need to manage the business effectively on both a daily operational basis and a strategic planning basis.

The reporting module must be capable of generating, on demand and without any manual data compilation, a comprehensive range of report types: daily sales summary reports presenting total revenue, transaction count, and service category breakdowns for any selected date; weekly and monthly financial statements aggregating daily performance data across the selected period with trend indicators; workstation utilization reports presenting occupancy rates, average session durations, and revenue contributions for each workstation; service category performance reports presenting volume and revenue metrics for each service type; staff performance reports presenting transaction volume and revenue data attributed to each staff member's sessions; and any additional operational analytics that management identifies as relevant to their specific decision-making needs. Each report must be generated directly and automatically from the database records created by the operational modules, with no human intermediation in the compilation process that could introduce errors or delays. The computational accuracy of the reports is therefore a direct function of the accuracy of the underlying transactional records — and because those records are generated automatically by the system's operational modules rather than entered manually, they reflect the actual operational events of the business with a fidelity that manual records cannot match.

The presentation format of the reports must prioritize readability and actionability for a management audience that may not have specialized financial training. Clear tabular layouts with appropriate column headers, totals, and summary lines; logical organization from highest-level summary to granular detail; and consistent formatting conventions across all report types — these design standards ensure that management users can extract the operational insights they need from reports without requiring interpretation assistance or specialized knowledge of how to read financial documents.

1.4 Scope and Delimitation

Scope

This study covers the design, development, and initial deployment of a dedicated branch management system tailored specifically for the operational needs of Q-Solutions business centers. The system is built to streamline and digitize the day-to-day service workflows that staff and management handle across branches.

The following functional areas fall within the scope of this study:

PC Time-Tracking Module: The system will provide staff with the ability to start and stop session timers for individual computer units. It will automatically calculate the duration of each session and compute the corresponding charges based on predefined rates. The time-tracking module will maintain a real-time view of active and idle units, allowing staff to monitor usage across all stations at a glance.

Service Categorization and Transaction Logging: Beyond PC rental, the system will support the recording and categorization of other common shop services such as printing, photocopying, scanning, and encoding. Each transaction will be logged with relevant details including service type, quantity, unit price, and the computed total, enabling accurate per-transaction records.

Automated Receipt Generation: Upon the completion of a transaction or session, the system will automatically generate a structured receipt summarizing the services availed, the duration of use where applicable, the itemized costs, and the total amount due. This feature eliminates manual computation and reduces human error in billing.

Inventory and Service Rate Management:: The system will allow authorized personnel to configure and update service rates, manage available service types, and maintain basic records for consumable supplies such as paper and ink. This ensures that billing calculations remain consistent and up to date.

User Access and Role Management: The system will implement basic role-based access control, distinguishing between staff-level and administrator-level accounts. Administrators will

have access to configuration settings, transaction history, and summary reports, while staff will be limited to transaction processing and session management.

Reporting and Transaction History: The system will generate daily, weekly, and monthly summaries of transactions, service usage, and revenue. These reports are intended to support management in reviewing branch performance and making informed operational decisions.

Local Area Network (LAN) Deployment: The system is designed to operate within the internal network of each Q-Solutions branch. All data will be stored and processed locally on a server machine accessible to connected workstations within the same LAN environment, ensuring stable performance without dependency on external internet connectivity.

Delimitation

While the system addresses the core operational needs of Q-Solutions branches, several areas are intentionally excluded from the current scope in order to maintain a focused and achievable development objective.

No Online or Remote Access: The system will not support access from outside the branch network. Remote monitoring, cloud-based data synchronization, or web-based dashboards accessible over the internet are not included in this iteration of the system.

No Customer-Facing Application: The system is intended exclusively for internal use by branch staff and management. There will be no customer-side portal, self-service kiosk interface, or mobile application through which customers can independently book sessions, check availability, or manage their accounts.

No Online Booking or Reservation System: Features for advance reservations or appointment scheduling via external platforms or messaging integrations are outside the scope of this study.

No Multi-Branch Centralized Network: While the system may be deployed across multiple branches individually, centralized real-time synchronization between branches over a wide area network (WAN) is not covered. Each branch will operate its own independent local instance of the system.

No Payment Gateway Integration: The system will not integrate with digital payment processors, e-wallets, or point-of-sale (POS) hardware for automated payment collection. Billing and payment recording will be handled manually by staff using the generated receipt as reference.

No Biometric or RFID Integration: Customer identification features such as loyalty card scanning, RFID-based logins, or biometric authentication are not within the scope of this study.

These delimitations reflect the current phase of the system's development and the practical constraints of the project. Features excluded at this stage may be considered for future versions or subsequent research studies as the operational needs of Q-Solutions evolve.

1.5 Significance of the Study

Q-Solutions Management and Ownership - For the ownership and management of Q-Solutions, the significance of this study is most directly and concretely expressed in the operational control, financial visibility, and strategic decision-making capability that the proposed system delivers. The transition from a paper-based, manually managed operational framework to a real-time, database-driven management platform represents a qualitative transformation in the quality of management that is possible within the business — not merely an incremental improvement in existing processes, but the creation of fundamentally new management capabilities that the current system structurally cannot provide.

The financial dimension of this transformation is perhaps the most immediately quantifiable. The elimination of revenue leakage through automated session tracking — ensuring that every session's duration is accurately recorded and every session extension is promptly identified and

charged — directly increases revenue collection relative to the current manual system. The elimination of billing computation errors through automated calculation ensures that every transaction is billed correctly, eliminating both the undercharging errors that reduce revenue and the overcharging errors that damage customer relationships. The comprehensive transaction records generated by the automated system provide the audit trail that allows management to detect and investigate any remaining discrepancies, creating an accountability environment that further protects financial integrity.

Beyond the immediate financial benefits, the system's reporting and analytics capabilities transform the management information landscape of the business. Where previously management operated with delayed, incomplete, and inaccurate financial summaries derived from manual record compilation, the automated system provides on-demand access to accurate, comprehensive, real-time financial data. This transformation in information quality enables a corresponding transformation in management quality — the shift from reactive, retrospective management (responding to problems identified after the fact in periodic summaries) to proactive, data-driven management (identifying and responding to operational patterns and emerging issues in real time, on the basis of accurate current data).

Staff - For the cashiers and attendants who constitute Q-Solutions' frontline operational staff, the significance of the system is experienced most directly as a reduction in the cognitive burden and operational stress associated with manual administrative management. The elimination of the need to simultaneously monitor multiple session timers, compute billing totals under time pressure, and maintain accurate paper records while managing customer interactions removes the most cognitively demanding and error-prone components of the staff role — freeing attention, mental capacity, and time for the customer service functions that require genuine human judgment, interpersonal skill, and operational expertise.

This cognitive load reduction is not merely a comfort improvement — it has direct operational consequences. Staff who are not under cognitive overload from manual administrative tasks perform all of their other operational functions better: they provide more attentive customer service, make fewer errors in the tasks that remain human-mediated, and are more available to

respond to unexpected operational situations that require flexible, judgment-based responses. The reduction in error-driven stress — the anxiety of knowing that manual billing computations and timing estimates are prone to error, and the discomfort of managing customer disputes about billing accuracy — also contributes to a more positive working environment that may improve staff satisfaction, reduce turnover, and enable better recruitment of capable individuals for operational roles.

The accountability dimension of the system's design also benefits staff, though this benefit is sometimes less immediately visible. Staff members who process transactions through the automated system have a complete, timestamped, system-generated record of every action they take — every session they initiate, every charge they apply, every payment they process. This record protects them from false accusations of billing error or revenue misappropriation, providing the objective evidence that demonstrates the correctness of their actions in any dispute or investigation. The system thus serves not only as a management oversight tool but as a staff protection tool, creating a fair and transparent accountability environment for everyone who uses it.

Customers - For the customers of Q-Solutions, the significance of the system is centered on two primary improvements in service quality: greater transparency and accuracy in billing, and faster, more attentive service delivery. These improvements address the specific customer experience failures that manual management creates and that Villanueva and Mercado (2022) documented as the primary drivers of customer dissatisfaction in manually managed Philippine internet café operations.

The billing transparency delivered by the system's itemized transaction records — automatically generated documents that clearly show every service consumed, the applicable rate, and the computed charge — gives customers objective, system-generated evidence of the accuracy and fairness of their bills. This transparency directly addresses the trust deficit that manually computed bills create, because it replaces a staff member's manual calculation (which customers have no basis to independently verify) with a system-generated record (which customers can review and verify against their own understanding of what they consumed). Customers who trust

the billing process are more satisfied with their overall service experience, more likely to return for future visits, and more likely to recommend the establishment to others.

The operational efficiency improvements delivered by the system — faster checkout processing, more responsive staff attention, and better-organized operations generally — translate directly into improved service quality at the point of customer interaction. Customers who receive prompt attention, efficient service processing, and accurate billing experience a service quality level that reflects positively on the establishment and supports the customer loyalty and positive word-of-mouth that are among the most important competitive assets a community-focused small business can develop.

BSIT Students - For the academic and research community, this study makes several distinct and complementary contributions to the literature on information systems for small and medium enterprises in developing-world contexts.

The study contributes a rigorously documented case study of a complete automated management system development lifecycle — from problem identification and requirements analysis through system design, development, testing, deployment, and evaluation — in the specific operational and cultural context of a Philippine micro-enterprise. This documentation provides an empirically grounded reference for subsequent researchers working on related applied systems development problems, reducing the knowledge gap between theoretical frameworks for information system design and the practical realities of implementation in resource-constrained, developing-world deployment contexts.

The study contributes specifically to the sub-field literature on management systems for multi-service digital service establishments — a category of business that has received less systematic research attention than single-service internet cafés despite its growing prevalence, operational complexity, and social significance in developing-world communities. The architectural and design challenges associated with managing multiple concurrent service types — PC rental, printing, photocopying — within a unified automated platform are documented in sufficient technical detail to advance the collective understanding of this problem domain and to provide a practical architectural reference for future developers.

The study also contributes to the literature on technology adoption and organizational transformation in Philippine SME contexts — documenting not only the technical design of the proposed system but the organizational conditions, implementation challenges, and user behavior factors that influence the practical effectiveness of technology-driven operational transformation in this specific context.

Software Developers and Technology Practitioners - For software developers and technology practitioners working in small business automation contexts — particularly those working with clients in developing-world markets where cost constraints and infrastructure limitations shape the feasible solution space — this study provides a detailed, practical reference for the design and implementation of automated management systems using open-source web technologies in resource-constrained deployment environments.

The system's reliance on PHP and MySQL — technologies that are universally available, comprehensively documented, supported by large developer communities, and maintainable without specialized expertise or significant licensing costs — demonstrates the feasibility of delivering effective small business automation within the tight cost constraints of micro-enterprise clients who cannot afford enterprise-grade technology solutions. The design patterns, database schema architectures, timer synchronization approaches, and reporting query strategies documented in this study represent practical, tested solutions to common problems in small business management system development that future developers can adapt and apply in analogous contexts.

1.6 Definition of Terms

Automated Management System

Conceptual: An automated management system is a technology platform that integrates software components to execute, coordinate, and monitor defined enterprise workflows without requiring continuous manual intervention at each step of each process. Automation in this context refers to the system's capacity to perform specified operations — computation, data recording, status display, report generation — in response to defined triggers, applying predetermined rules to data

drawn from a central repository, without requiring a human operator to manually execute each computational or data management step.

Operational Definition: Within this study, the term refers specifically to the PHP and MySQL web application developed as the primary deliverable of the research — the integrated system that automates the session management, billing computation, record keeping, and financial reporting functions currently performed manually at Q-Solutions. This application constitutes an automated management system in the conceptual sense because it performs these functions through software execution of predefined logic operating on database-stored data, eliminating the human-executed manual processes that currently implement these functions in the establishment's operations.

PC Rental Monitoring

Conceptual Definition: PC rental monitoring encompasses the structured processes of tracking the utilization of computer workstations as rented commercial resources — recording the temporal boundaries of each rental session, measuring and displaying elapsed session duration, managing the transitions between session states (available, occupied, extended, concluding), and maintaining comprehensive records of workstation utilization for billing and operational analysis purposes.

Operational Definition: Within this study, the term refers specifically to the timer-based session tracking functionality of the proposed system — the module that maintains server-side records of the status of every workstation in the Q-Solutions inventory and, for each occupied workstation, precisely tracks the duration of the active session from its database-recorded initiation timestamp, displaying continuously updated status information on the administrative dashboard and generating alerts when configured duration thresholds are reached.

Analog Operations

Conceptual Definition: Analog operations refers to administrative and operational frameworks that rely entirely on physical media, non-digital instruments, and manual human processes to

execute business functions. The term distinguishes pre-digital management methods — handwritten records, paper-based document systems, manual arithmetic, mechanical timing devices — from digital alternatives that automate these functions through software and electronic data processing.

Operational Definition: Within this study, the term describes the current administrative framework of Q-Solutions — specifically, the handwritten session logbooks, wall clock and personal device timing methods, handheld calculator billing computation, and paper receipt documentation that together constitute the business's current operational management approach. These analog components are the specific targets of the proposed system's automation, and the documented inadequacies of these components in the Q-Solutions operational context constitute the primary justification for the system's development.

Revenue Leakage

Conceptual Definition: Revenue leakage describes the unrecovered loss of income that occurs when a business delivers services for which it fails to collect full payment, due to procedural failures, system vulnerabilities, human error, or deliberate misconduct. Revenue leakage is distinguished from intentional discounting or promotional pricing, which represent deliberate management decisions to accept below-standard rates for defined strategic purposes.

Operational Definition: Within this study, the term describes the specific financial losses documented at Q-Solutions as consequences of its manual administrative framework — sessions running beyond purchased durations without extension charges being collected, print jobs delivered without being recorded in billing computations, arithmetic errors producing understated transaction totals, and sessions initiated without being entered in the logbook. The quantification and elimination of revenue leakage through automated tracking and billing is one of the primary financial benefits the proposed system is designed to deliver, and the reduction in revenue leakage is one of the primary metrics against which the system's effectiveness will be evaluated.

Real-Time Tracking

Conceptual Definition: Real-time tracking refers to the continuous, instantaneous recording and synchronization of operational data with a central repository, such that the data available for query and display at any given moment accurately reflects the current operational state of the system without meaningful delay. Real-time systems are distinguished from batch processing systems, which accumulate data over defined periods and process it collectively at scheduled intervals rather than continuously as events occur.

Operational Definition: Within this study, real-time tracking defines the core behavioral requirement of the session management module — the continuous updating of session duration data, workstation status information, and financial accrual calculations in the database and dashboard display, at sub-second intervals, ensuring that the administrative dashboard always presents an accurate, current picture of the establishment's operational state without requiring manual data refresh or re-entry by staff.

Database Normalization

Conceptual Definition: Database normalization is the process of designing a relational database schema to minimize data redundancy and maximize data integrity by organizing data according to formal structural rules — the normal forms of relational database theory — that ensure each data fact is stored in exactly one place and all relationships between data entities are represented through clearly defined relational structures rather than through data duplication.

Operational Definition: Within this study, normalization refers to the structural design principles applied in the construction of the MySQL database schema underlying the proposed system — specifically, the application of third normal form standards to ensure that each entity in the Q-Solutions operational domain (sessions, transactions, workstations, staff members, service

types, pricing configurations) is represented in the most appropriate relational structure, with inter-entity relationships maintained through properly implemented foreign key constraints that enforce referential integrity across the complete dataset.

Session Management

Conceptual Definition: Session management encompasses the complete operational lifecycle of a customer's rental resource engagement — the sequence of states from resource availability through session initiation, active use, any extensions, and session conclusion to checkout settlement. Comprehensive session management requires accurate tracking of state transitions, reliable measurement of session duration, appropriate handling of lifecycle events including extensions and early terminations, and the generation of complete records sufficient for billing and audit.

Operational Definition: Within this study, session management refers to the complete set of system functions governing the lifecycle of PC rental sessions at Q-Solutions — including the initiation of database session records at workstation assignment, the continuous server-side timing throughout the session, the processing of extension events, the generation of expiration alerts, the initiation of checkout processing at session conclusion, and the finalization of session records with complete event timing logs.

Role-Based Access Control

Conceptual Definition: Role-based access control is a security architecture principle through which system permissions are assigned to defined organizational roles rather than individual user accounts, such that the set of system functions available to any individual user is determined by their assigned role rather than by individually configured permissions. This approach simplifies permission management, enforces consistent access boundaries across users in the same organizational role, and aligns system access boundaries with organizational authority structures.

Operational Definition: Within this study, role-based access control refers to the two-tier permission architecture implemented in the Q-Solutions management system. The Administrator

role provides access to the complete set of system functions including operational management, system configuration, user account administration, historical data review, and reporting. The Staff role provides access to the operational transaction-processing subset of functions — session management, service recording, and checkout processing — while restricting access to configuration, historical modification, and administrative management functions outside the operational scope of the Staff role.

Billing Computation Engine

Conceptual Definition: A billing computation engine is a software module that automatically calculates charges for delivered services by applying configured pricing parameters to service consumption data, producing accurate transaction totals without requiring manual arithmetic input. The engine's accuracy derives from the precision of its configured parameters and the correctness of its computational logic rather than from human calculation, making it immune to the arithmetic errors, cognitive load effects, and inconsistency factors that affect manual billing.

Operational Definition: Within this study, the billing computation engine refers to the PHP software module that calculates the total charges for Q-Solutions customer transactions by automatically applying the business's configured pricing rules — rental rates, print job rates, photocopy rates — to the service consumption data recorded by the session tracking and service logging modules, generating itemized transaction records that present the complete, transparent breakdown of each transaction's charges.

1.7 Conceptual Framework

1.7.1 The Input-Process-Output Model as Analytical Framework

The development, deployment, and evaluation of the Automated Sales Monitoring and Resource Management System for Q-Solutions is organized within the Input-Process-Output analytical framework — a systemic model widely applied in information systems design, engineering project management, and organizational research to represent the logical architecture of a designed system in terms of the preconditions it requires, the transformative processes it encompasses, and the outcomes it produces.

The IPO model has particular utility as the conceptual framework for this study because it makes explicit the three-way relationship between what is needed (inputs), what is done (processes), and what results (outputs) at each stage of the system's lifecycle. Applied to a technology development and implementation project, the model organizes the full complexity of the research into a logical structure that clearly shows how operational knowledge and technical specifications are transformed into a working system through engineering processes, and how that system transforms operational realities through its deployment and use. This clarity of logical structure is valuable not only as an analytical tool for the researchers but as a communication tool that allows stakeholders — management, staff, and academic readers — to understand the system's design rationale and expected impact at an appropriate level of abstraction.

The IPO framework as applied in this study encompasses three phases, each internally differentiated into multiple components that reflect the specific structure of the Q-Solutions operational context and the technical characteristics of the proposed system.

Input Phase: Foundational Requirements and System Preconditions

The input phase encompasses all the foundational knowledge, operational data, technical specifications, and contextual understanding that must be gathered, documented, and translated into system design specifications before development work can begin. The inputs are organized into four primary categories.

Documented Operational Workflows and Administrative Requirements

The most fundamental category of inputs is the comprehensive documentation of Q-Solutions' current operational processes — the specific sequences of administrative activities associated with each type of customer transaction, the specific roles and task responsibilities of different staff categories, the staffing patterns and peak demand periods that characterize the business's operational tempo, and the specific operational pain points that the system must address.

This documentation is gathered through a combination of direct operational observation, structured interviews with management and staff, and analysis of existing paper records. Operational observation during both peak and off-peak periods provides data about the actual

pace and complexity of the operational environment and the specific conditions under which manual management failures occur most frequently. Management interviews provide insight into the strategic priorities for the system and the specific features that management considers most critical for operational improvement. Staff interviews provide practical insight into the specific difficulties of the current manual processes from the perspective of the operators who experience those difficulties most directly. Together, these data sources constitute a comprehensive operational knowledge base that informs every aspect of the system's design.

Service Pricing Architecture and Billing Rules

The complete pricing structure for all Q-Solutions service offerings constitutes a critical category of technical input that determines the computational architecture of the billing engine. This input encompasses the hourly rental rate for PC stations, any time-of-day or customer category variations in that rate, the complete matrix of printing rates across all paper size and ink type combinations the business supports, the per-copy photocopying rate, and any promotional discount rules or special pricing programs currently in effect.

Beyond the simple enumeration of current rates, this input requires a structural analysis of the pricing architecture — the number of distinct pricing variables, the logical conditions under which different rates apply, the computational sequence required to apply these rules correctly to a multi-service transaction, and the frequency with which pricing parameters change over time. This structural analysis informs the design of both the billing computation algorithm and the pricing configuration management interface.

Technical Infrastructure and Deployment Environment

The hardware and software specifications of the Q-Solutions technical environment — server hardware capabilities, client workstation configurations, browser software versions, local area network architecture, and existing system software — constitute the technical deployment context within which the system must operate. These specifications define the performance constraints within which the application must function, the compatibility requirements that the web interface must meet, and the integration constraints that the application must respect.

Documenting these specifications requires systematic assessment of the existing infrastructure, producing a technical baseline that ensures the system is designed to operate correctly within the actual deployment environment rather than against an idealized technical configuration that may not reflect reality.

User Proficiency Profile and Training Requirements

The technical proficiency and operational experience of the staff members who will use the system constitutes a human factors input that directly shapes the interface design, training program, and implementation approach of the system. Staff members who are comfortable with web-based applications require different interface design accommodations than those who are new to web-based tools. The specific operational challenges that staff face in the current system — the timing difficulties, the billing uncertainties, the record-keeping burdens — inform the specific interface design choices that will make the new system feel like a solution rather than an additional burden.

Process Phase: System Engineering Methodology

The process phase encompasses the complete sequence of engineering activities through which the inputs of the previous phase are transformed into the deployed, evaluated system. This transformation proceeds through five interdependent stages.

Requirements Analysis and Formal Specification

Requirements analysis translates the operational knowledge documented in the input phase into a formal technical specification — the complete, unambiguous statement of what the system must do (functional requirements), how well it must do it (non-functional performance requirements), and within what boundaries it must operate (system constraints).

The formal specification produced by this stage serves as the primary reference document for all subsequent design and development activities, providing the standard against which every design decision is evaluated and the criteria against which the completed system is tested.

The requirements specification disciplines the translation between business language and technical language, ensuring that the operational needs identified through observation and interviews are accurately and completely captured as technical requirements that can be implemented in code and verified through testing. Gaps or ambiguities in the requirements specification become errors or inconsistencies in the implemented system; therefore, the rigor and completeness of this stage is a critical determinant of the overall quality of the development process.

Architectural and Detailed System Design

The system design stage transforms the requirements specification into the architectural blueprint from which the system will be built. Design artifacts produced in this stage include the entity-relationship diagram for the database schema, the application architecture diagram showing the relationships between system components and modules, interface wireframes for all dashboard screens and user workflows, data flow diagrams showing how information moves through the system, and algorithmic pseudocode for the key computational modules — particularly the timer synchronization logic and the billing computation engine.

The design stage is where the fundamental architectural decisions that shape the system's behavior are made — the client-server interaction model, the timer state authority architecture, the database normalization approach, the reporting query design. Each of these decisions is documented with explicit justification connecting the design choice to the requirements it satisfies and the architectural principles it applies.

Iterative Coding and Module Development

The development stage implements the system design in working software, proceeding through an iterative cycle of coding, unit testing, and refinement. Backend application logic is implemented in PHP following a modular architecture that separates functional concerns — session management, billing computation, reporting, user authentication, system configuration — into distinct modules with clearly defined interfaces. Frontend interface components are implemented in HTML, CSS, and JavaScript following the wireframe specifications. Database

objects — tables, relationships, indexes, stored procedures — are implemented in MySQL following the normalized schema design.

The iterative development approach — in which individual modules are developed, tested, and refined before integration — reduces the risk of accumulated errors and allows systematic validation of each component's correctness before it is incorporated into the integrated system.

Integration, Verification, and Validation Testing

The testing phase constitutes a comprehensive quality assurance methodology designed to evaluate system correctness, performance, and compliance against functional and non-functional requirements. This process follows a structured, multi-tier testing program:

- **Unit Testing:** Focuses on isolated components, verifying that individual modules, algorithms, and logic units execute correctly prior to integration.
- **Integration Testing:** Evaluates inter-module communication, API endpoints, and data interfaces to ensure that disparate components function cohesively as a unified platform.
- **System Testing:** Conducts end-to-end evaluation of the fully integrated software under realistic, operational scenarios to validate end-to-end business workflows.
- **Performance Testing:** Simulates high concurrency, peak load conditions, and stress thresholds to measure latency, resource consumption, and overall system stability.
- **Security Testing:** Assesses access control policies, role-based authorization rules, authentication mechanisms, and vulnerability vectors to prevent unauthorized data access.
- **User Acceptance Testing (UAT):** Involves end-user participation within a simulated operational environment to validate workflow alignment, business logic correctness, and overall usability.

All execution runs, anomaly reports, and test results are logged within a defect tracking system. Issues undergo structured resolution and re-testing cycles until every functional requirement is fully verified. The final output of this stage is a quality assurance report establishing deployment readiness.

Deployment, Training, and Controlled Prototype Evaluation

The deployment phase transitions the validated software into the operational environment of Q-Solutions. The execution plan encompasses installation, parameter configuration, capacity building, and empirical evaluation:

1. **Environment Provisioning and Configuration:** The software is deployed onto local server infrastructure, configured with actual operational parameters, local database connections, billing models, pricing matrices, and workstation inventories.
2. **Staff Training and Onboarding:** Structured training sessions are delivered to operational staff and administrative personnel to build system proficiency, standardize operational procedures, and reduce user error during data entry.
3. **Controlled Prototype Operation:** A monitored pilot period is established where the system operates in live operational conditions.
4. **Empirical Evaluation Framework:** Quantitative and qualitative data points are captured systematically throughout the pilot run:
 - Accuracy Metrics: Error rate reductions, billing precision, and inventory tracking fidelity.
 - Efficiency Metrics: Transaction processing speed, resource turnover rates, and task duration comparisons.

- Usability Assessments: User satisfaction indicators, task completion times, and system learnability feedback.

The data gathered during this evaluation provides the evidence base required to quantify operational efficacy and draw analytical conclusions regarding system performance.

Output Phase: System Deliverables and Operational Impact

Primary Technical Deliverable: The Integrated System

The primary deliverable of this research is the fully functional Automated Sales Monitoring and Resource Management System. Deployed as a web application running within the Q-Solutions Local Area Network (LAN), the platform integrates five core modules into a single interface:

[Central Web Platform]

- |
- |— Module 1: Authentication & Access Control
- |— Module 2: Workstation & Resource Tracking
- |— Module 3: Automated Billing & Transaction Processing
- |— Module 4: Sales Monitoring & Financial Reporting
- |— Module 5: System Administration & Inventory Management

This deployed platform serves as both the technical deliverable of the software engineering process and the primary subject of study for empirical performance evaluation.

Operational Improvement Outcomes

The system's deployment delivers measurable operational advancements for Q-Solutions, establishing the business value of the software. Key operational metrics evaluated include:

- Billing Accuracy & Revenue Assurance: Eliminates manual calculation errors, reduces revenue leakage, and ensures consistent rate enforcement.

- **Transaction Processing Speed:** Accelerates customer check-in, check-out, and payment workflows to minimize wait times.
- **Operational Capacity:** Increases throughput per staff member by automating routine administrative tasks and resource logging.
- **Data Visibility & Reporting:** Provides real-time administrative dashboards for immediate access to sales figures, usage logs, and resource utilization metrics.

Documenting these metrics provides empirical proof that automated management technologies offer practical solutions for micro-enterprises operating in developing markets.

Academic Knowledge Contribution

Beyond operational improvements for Q-Solutions, this study contributes scholarly and practical insights to the broader software engineering and academic communities:

- **Empirical Performance Data:** Documents real-world performance metrics of web-based resource monitoring systems within bandwidth-constrained or localized network environments.
- **Applied Software Methodology:** Establishes a practical development framework for tailoring enterprise software solutions to micro-enterprises in developing economies.
- **Architectural Blueprints:** Provides open-reference system architecture, modular design patterns, and database schemas for developers and technology consultants addressing similar micro-business automation challenges.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

The global landscape of business management has been significantly transformed by the advent of automated systems, particularly in service-oriented industries. A growing body of international research affirms that the integration of technology in managing daily business operations leads to measurable improvements in efficiency, revenue accuracy, and customer satisfaction. The following foreign studies provide a broad foundation for understanding the necessity and impact of automated monitoring and billing systems in environments similar to Q-Solutions Business Center. The transition from manual to digital management has been documented across multiple continents and business types, and the consistency of findings across these varied contexts lends particular credibility to the conclusions drawn from individual studies. This section presents a thorough review of the most relevant and significant international research, organized thematically to highlight the interconnected dimensions of automated system adoption in service-oriented small and medium enterprises.

According to Williams and Thompson (2021), the integration of automated billing systems in service-oriented small businesses significantly reduces operational costs and improves customer trust. Their research, which examined over 200 small business hubs across three metropolitan areas in the United States, found that the shift from manual to automated billing resulted in an

average cost reduction of 22% over a 12-month period. This reduction was attributed primarily to the elimination of redundant administrative tasks, the minimization of billing disputes, and the reduction of time spent reconciling daily transaction records. Their study on "Smart Business Hubs" further highlighted that manual time-tracking leads to a 15% margin of error in revenue collection, a figure that compounded over months and years into substantial financial losses for business owners. The researchers conducted their investigation using a mixed-methods approach, combining quantitative analysis of financial records from participating businesses with qualitative interviews of owners, managers, and frontline staff members. The convergence of financial data and firsthand accounts created a particularly compelling evidentiary foundation for their conclusions.

Williams and Thompson (2021) further observed that customers expressed significantly higher confidence in automated receipts and timer-based billing systems compared to handwritten logs. This psychological dimension of trust is critical in service businesses where customers pay for time-based usage, as in computer rentals and printing services. When customers can visually observe a digital timer counting their session in real-time, the perceived fairness of the transaction increases, which in turn reduces disputes and improves repeat patronage. The researchers conducted a series of customer perception surveys both before and after the introduction of automated systems in a sample of ten business centers and found that the proportion of customers who rated the billing process as "fair" or "very fair" increased from 54% to 89% following system adoption. This 35-percentage-point improvement represents one of the most striking individual findings in the study and speaks directly to the customer-facing value of transparent, system-generated billing. This finding is particularly relevant to Q-Solutions, where transparency in billing directly affects customer loyalty in a competitive supermarket environment.

Williams and Thompson (2021) also devoted considerable attention to the operational workflow changes that accompanied system adoption in their sample businesses. They found that the introduction of automated billing systems prompted a broader reorganization of staff responsibilities, with employees freed from the constant mental task of tracking time and calculating charges being redeployed toward customer service, maintenance, and quality assurance functions. In several of the businesses studied, this reallocation of human effort led to

noticeable improvements in service turnaround times and a reduction in customer wait times during peak hours. The researchers argued that this secondary benefit of automation — the liberation of human capacity for higher-value activities — was frequently underestimated by business owners evaluating the return on investment of system adoption, who tended to focus narrowly on direct cost savings rather than the broader operational improvements enabled by the freeing of staff cognitive resources.

Complementing this perspective, Davis (2020) emphasized that centralized monitoring dashboards allow managers to oversee multiple service points simultaneously, thereby increasing overall organizational efficiency. In his study of franchise-based service centers in the United Kingdom, Davis documented how the implementation of a web-based monitoring interface enabled branch supervisors to track active sessions, monitor printer queues, and generate sales summaries without being physically present at each terminal. His research demonstrated that this capability reduced supervisory response time to service issues by 40%, as problems such as idle machines, unauthorized usage, or billing discrepancies could be flagged and addressed in real time. Davis's study is notable for its focus on the management dimension of automated systems, recognizing that the benefits of digitalization extend beyond the point of transaction to encompass the broader organizational challenge of effective oversight and control.

Davis (2020) also introduced the concept of "operational visibility," defining it as the degree to which a manager can access, interpret, and act upon business data at any given moment. He argued that traditional manual systems inherently limit operational visibility because data is fragmented across logbooks, receipts, and staff memories, all of which are susceptible to loss, alteration, or misinterpretation.

In contrast, a centralized digital dashboard consolidates all operational data into a single interface, enabling informed decision-making with minimal delay. Davis surveyed 78 branch managers across 15 franchise service center chains and found a strong positive correlation between the degree of operational visibility afforded by their management systems and their scores on composite measures of branch performance, including revenue consistency, customer satisfaction, and staff retention. This empirical link between system design and business performance outcomes provides a powerful argument for investing in high-visibility monitoring

infrastructure. For a business like Q-Solutions, which operates within a busy supermart branch and handles multiple service types simultaneously, operational visibility is not merely a convenience but a necessity for maintaining control over a complex, high-volume service environment.

Davis (2020) extended his analysis to consider the temporal dimension of operational visibility, noting that the value of real-time data access is qualitatively different from the value of retrospective reporting. While end-of-day sales summaries and weekly transaction reports provide valuable information for strategic planning and financial management, real-time dashboards enable a qualitatively different mode of management — one characterized by immediate situational awareness and the ability to intervene in developing problems before they escalate into significant disruptions. Davis cited several case studies in which real-time dashboard access allowed managers to detect and address equipment malfunctions, unauthorized service usage, and billing errors within minutes of their occurrence, preventing customer complaints and revenue losses that would have been inevitable had the issues only been discovered during end-of-day reconciliation.

Anderson and Reyes (2019) conducted a comparative study of analog versus digital service management systems in small businesses across Southeast Asia and Australia. Their findings revealed that businesses using digital systems reported a 30% decrease in customer complaints related to billing and a 25% improvement in daily revenue accuracy. The researchers attributed these improvements to the removal of human subjectivity from the billing process — when a timer controls the session, neither the staff member nor the customer can dispute the duration of service based on personal recollection. Anderson and Reyes recruited 120 small businesses for their study, randomly assigning half to receive automated system implementation while the other half continued with manual management, then tracked outcomes over a six-month period. This quasi-experimental design represented one of the most methodologically rigorous approaches in the literature reviewed for this study, lending particular credibility to the causal conclusions drawn by the researchers.

Anderson and Reyes (2019) also noted that digital systems enabled more sophisticated data analytics, allowing business owners to identify peak usage hours, most popular services, and seasonal demand patterns, all of which contributed to more strategic resource allocation. Business owners in the digital system group were significantly more likely to have made operational changes — adjustments to opening hours, staffing

schedules, pricing structures, or service offerings — in response to data insights during the study period, compared to their counterparts in the manual system group. This finding highlights an important distinction between operational benefits, which affect the efficiency of day-to-day service delivery, and strategic benefits, which affect the long-term direction and competitiveness of the business. The ability of an automated monitoring system to generate both types of benefit simultaneously represents a particularly compelling argument for its adoption in a business like Q-Solutions, where both operational efficiency and strategic agility are important to long-term success.

Kim and Park (2020) explored the adoption of point-of-sale (POS) integrated systems in Korean business centers and PC cafes, observing that the fusion of billing, inventory, and session management into a single platform eliminated the need for separate software tools and reduced staff training time by approximately 35%. Their study examined 60 business centers in Seoul and Busan, comparing establishments that used integrated unified systems against those that relied on a combination of separate tools for different operational functions. The researchers found that integration significantly reduced the frequency of inter-system data conflicts — discrepancies between sales logs, session records, and inventory counts that were common in multi-tool environments — and that these conflicts had been a significant source of both staff frustration and managerial distrust of data accuracy in the pre-integration period.

Kim and Park (2020) also examined the learning curve associated with integrated system adoption, finding that despite the broader functional scope of unified systems compared to individual tools, the overall training time required was significantly lower. The researchers attributed this counterintuitive finding to the elimination of context switching — the cognitive cost of moving between different software interfaces with different interaction patterns,

conventions, and data formats. When all functions are accessible within a single coherent interface, staff members develop competency more quickly and retain their skills more reliably than when they must master multiple disconnected tools. This finding has direct implications for the Q-Solutions system design, which prioritizes functional integration specifically to minimize training complexity and accelerate staff proficiency.

Foster and Chen (2022) investigated the role of automated receipt generation in customer service quality within business centers in Canada and Singapore. Their research found that the issuance of itemized digital receipts — detailing service type, duration, rate per unit, and total amount — significantly reduced post-transaction disputes and increased customer satisfaction scores by an average of 18 percentage points. The authors argued that receipt transparency functions as a form of non-verbal communication between the service provider and the customer, signaling professionalism, accountability, and institutional reliability.

Their study involved both quantitative analysis of dispute records and customer satisfaction surveys, and qualitative analysis of customer focus group discussions, revealing that customers interpreted the provision of detailed receipts as evidence that the business had nothing to hide and was committed to fair dealing.

Foster and Chen (2022) identified a particularly interesting dimension of receipt transparency in their focus group data: customers reported that the detail and clarity of a receipt influenced their perception not only of the specific transaction but of the business as a whole. A clear, itemized receipt did not merely resolve any uncertainty about the current charge; it generated what the researchers called a "halo effect" of institutional trust that predisposed customers to view the business favorably in subsequent interactions. This halo effect was found to be particularly strong among first-time customers, who had no prior relationship with the business on which to base their trust judgments and therefore relied heavily on visible signals of professionalism and accountability. The implication for Q-Solutions is significant: the quality of the automated receipts generated by the monitoring system is not merely an administrative detail but a front-line customer service and brand-building tool.

Nguyen (2021) studied the implementation of cloud-based and local network management tools in Vietnamese internet cafes and found that even low-cost local network solutions, when designed with clear user interfaces and automated timing functions, could replicate the core benefits of enterprise-grade systems at a fraction of the cost. His research examined 35 internet cafes across Ho Chi Minh City and Hanoi, comparing the operational performance of establishments using different levels of system sophistication, from basic timer software to fully integrated cloud-based management platforms. Contrary to the expectation that more sophisticated and expensive systems would uniformly outperform simpler alternatives, Nguyen found that the most important determinant of system effectiveness was not technical sophistication but fit between the system's design and the specific operational requirements of the business.

Nguyen (2021) argued that the most effective systems are those tailored to the specific operational context rather than adapted from large-scale corporate platforms designed for different environments and user bases. He documented several cases in which cafes had invested significantly in enterprise-grade systems only to find that the complexity and overhead of these systems created new operational challenges that offset the benefits of their advanced features. Conversely, cafes that had invested in simpler, purpose-built solutions reported high levels of user satisfaction and consistent operational improvements even without the advanced analytics and multi-branch management features of more expensive alternatives.

This finding validates the approach taken in the development of the Q-Solutions system, which was designed from the ground up to address the unique service mix and operational scale of its target environment rather than adapted from a generic business management platform.

Martinez and Liu (2023) provided more recent evidence supporting the shift toward automated management in multi-service business environments. Their longitudinal study, which tracked 15 business centers over a period of three years following system digitalization, found that

automated centers consistently outperformed their manual counterparts across a range of performance metrics including revenue consistency, customer retention, and employee satisfaction. The longitudinal design of their study is particularly valuable because it captures the trajectory of system impact over time, revealing patterns that cross-sectional studies cannot detect. Martinez and Liu found that the benefits of automation were not evenly distributed across time: operational improvements in accuracy and efficiency were evident within the first month of adoption, while strategic benefits in the form of data-driven decision-making and customer loyalty improvements became more pronounced over the first year and continued to strengthen through the second and third years of the study period.

Martinez and Liu (2023) also found that employees in automated environments reported lower levels of work-related stress, as the system handled the cognitive burden of timing, billing, and record-keeping, freeing staff to focus on customer interaction and service quality. This human-centric benefit of automation is often overlooked in purely financial analyses of system adoption but represents a significant dimension of operational improvement. The researchers used validated workplace stress measurement instruments and found that mean stress scores among business center staff decreased by 28% over the two years following system adoption, with the largest reductions occurring in response to the elimination of billing dispute situations, which were identified as the single most stressful recurring event in the pre-automation work environment.

Patel and Okonkwo (2022) examined the cybersecurity and data integrity dimensions of automated billing systems in service SMEs across India and Nigeria. Their research found that RDBMS-based billing systems, when properly implemented with appropriate access controls and audit logging, provided a level of transaction integrity that manual systems inherently cannot offer. The use of timestamped database entries, user authentication logs, and automated backup functions ensured that transaction records were verifiable and tamper-resistant.

The authors noted that the importance of this data integrity function was heightened in regulatory environments characterized by increasing tax authority scrutiny of SME financial records, a trend that they documented across both study countries and which has clear parallels in the Philippine regulatory context.

Patel and Okonkwo (2022) also explored the internal governance dimension of transaction integrity, noting that tamper-resistant records not only satisfy external regulatory requirements but also protect business owners against internal fraud and misappropriation. In their sample of 50 small service businesses, they found that 18% had experienced confirmed incidents of staff manipulation of manual billing records, and that an additional 24% suspected such manipulation but could not confirm it due to the inherent ambiguity of handwritten records. The introduction of automated systems with user authentication and audit logging effectively eliminated this category of internal fraud risk, as all transactions were linked to specific user accounts and any modification of records would be logged and attributable. This finding adds an important dimension to the business case for automated billing systems that extends beyond external customer relations to encompass internal organizational integrity.

Harrison and Montgomery (2021) examined the relationship between digital system adoption and business longevity in SME service businesses across the United Kingdom and Germany. Their study found that businesses that had adopted integrated digital management systems were significantly more likely to be operational five years after the study baseline compared to those that continued with manual management, even after controlling for other factors such as business age, location, and owner experience. The researchers argued that this longevity advantage was attributable to several interrelated factors: better financial performance due to reduced revenue leakage, better customer retention due to billing transparency, better management decision-making due to data availability, and better regulatory compliance due to accurate and auditable transaction records. This finding situates the adoption of automated monitoring systems not merely as an operational improvement but as a strategic investment in business sustainability.

Zhao and Williams (2020) investigated the specific challenges of managing multiple concurrent service types within a single business establishment, a challenge directly analogous to that faced by Q-Solutions. Their study, conducted in multi-service business hubs in Shanghai and London, found that the complexity of simultaneous service management grew non-linearly with the number of service types: managing two service types simultaneously was significantly more than twice as difficult as managing one, due to the cognitive overhead of context switching and the increased potential for inter-service billing errors. The researchers found that unified integrated systems reduced this complexity dramatically by creating a single unified view of all active

services, eliminating the need for staff to mentally integrate information from multiple separate records when handling

customers who used more than one service type during a single visit. Collectively, these foreign studies establish a compelling international consensus: automated, integrated monitoring systems represent the superior approach to managing time-based and transactional services in small to medium business environments. The convergence of their findings across different geographic, cultural, and economic contexts — from the United States and United Kingdom to South Korea, Vietnam, India, Nigeria, China, Australia, Canada, Singapore, and the Philippines — strengthens the case for the development and implementation of the Q-Solutions Business Center Monitoring System, which draws upon these established principles while adapting them to a highly specific local environment and a unique multi-service operational context.

2.2 Local Studies

The Philippine business landscape, particularly in the realm of small-to-medium enterprises offering internet, printing, and related services, has been the subject of a growing number of local academic studies that highlight the persistent challenges associated with manual management systems and the increasing imperative to adopt digital alternatives. These local studies provide context that is directly applicable to the operational environment of Q-Solutions Business Center, illuminating both the problems the system seeks to solve and the solutions that have proven effective in comparable Philippine settings. The following review covers the most relevant and significant Philippine studies, organized to progress from the broadest contextual findings to those most specifically relevant to the Q-Solutions operational environment.

In a study conducted by Villanueva and Mercado (2022) regarding Internet Cafe management in urban areas of the Philippines, it was found that manual logging is the primary cause of conflict between staff and customers due to perceived overcharging. The researchers surveyed 45 internet cafe operators and over 300 customers across three major Philippine cities, finding that 62% of customer complaints related to billing were rooted not in actual overcharging but in the absence of a verifiable, objective record of service duration. Customers who were presented only with a

handwritten log or a verbal declaration of time used were significantly more likely to dispute the charge, even when the staff's record was accurate. This finding reveals a fundamental limitation of manual billing systems that goes beyond simple inaccuracy: even when manual records are correct, their inherent unverifiability makes them insufficient to establish customer confidence in the fairness of the transaction.

Villanueva and Mercado (2022) conducted detailed case studies of five internet cafes that had transitioned from manual to automated timer-based billing systems during the study period, enabling them to compare operational data from before and after system adoption in the same establishments. These within-establishment comparisons are particularly valuable because they control for establishment-specific factors that might confound between-establishment comparisons, such as location, customer demographics, and owner management style. The case study data revealed that billing disputes, measured as instances of customer-initiated challenges to a bill that required staff intervention to resolve, decreased by 58% in the three months following system adoption compared to the three months preceding it. Customer satisfaction scores, measured through brief exit surveys, improved by an average of 22 points on a 100-point scale over the same period.

Villanueva and Mercado (2022) further documented that locally developed Timer Systems have been proven to enhance the transparency of transactions in Philippine internet cafes. These systems, though varying in sophistication from simple countdown displays to full integrated billing platforms, shared the common feature of displaying a countdown or elapsed-time indicator visible to both staff and customers. Beyond reducing disputes, the visible timer was found to have an important behavioral effect on customers: the ability to monitor their remaining session time allowed customers to manage their usage more effectively, reducing the frustration of unexpected session terminations and improving the overall service experience. Staff members also reported that the presence of a visible, system-controlled timer eliminated the awkward social dynamic of having to inform a customer that their time had expired, as the system itself served this function objectively and impersonally.

Garcia (2021) noted that small-to-medium enterprises in Iloilo are increasingly adopting digital logging systems to comply with more rigorous auditing and tax requirements imposed by the Bureau of Internal Revenue and local government units. His study, which focused on 30 SMEs across the service sector in Iloilo City, found that the BIR's increasing emphasis on digital record-keeping and the mandatory issuance of official receipts was driving a wave of system modernization among local business owners. Garcia found that businesses which had adopted digital transaction logging were significantly better prepared for BIR audits, with the ability to generate complete, date-stamped transaction histories at any point, compared to their manually operated counterparts who frequently reported incomplete or illegible logbooks during audit periods.

Garcia (2021) also documented that several Iloilo SMEs had faced penalties and back-tax assessments as a result of inadequate record-keeping, with the inability to produce complete and verifiable transaction records during BIR audits being cited as the primary reason for adverse audit outcomes. The financial impact of these penalties, combined with the reputational damage of being found non-compliant, provided a powerful motivating force for system adoption that operated independently of the operational efficiency arguments more commonly emphasized in the literature. Garcia argued that compliance risk represented an underappreciated dimension of the business case for digital system adoption among Philippine SMEs, particularly as regulatory scrutiny of the sector continued to intensify.

This regulatory dimension is directly applicable to Q-Solutions, which operates as a formal business establishment subject to all applicable BIR and local government requirements.

Garcia (2021) also examined the broader economic impact of digital adoption among Iloilo SMEs, finding that businesses that digitalized their transaction logging reported an average revenue increase of 12% in the year following adoption. This increase was attributed to several factors: the elimination of unrecorded transactions, the ability to analyze sales data and identify underperforming services, and the reduction of time spent on daily reconciliation, which freed staff to attend to more customers during peak hours. Garcia supplemented his quantitative analysis with in-depth interviews of ten business owners who had recently completed digital

system transitions, and these qualitative accounts provided rich contextual detail about the mechanisms through which digitalization generated revenue improvements. Several owners described discovering, through their new transaction logs, patterns of unrecorded or undercharged transactions that had been occurring systematically but had been invisible under manual systems.

Santos and Bautista (2020) conducted a study of computer rental businesses in Metro Manila, examining the operational challenges faced by small operators managing both computer rentals and auxiliary services such as printing and photocopying. Their research found that the simultaneous management of multiple service types within a single establishment created significant operational complexity when handled manually. Staff members were frequently required to track multiple computer session timers simultaneously while also monitoring print jobs, processing photocopying requests, and managing cash transactions, a combination of demands that consistently exceeded the attentional capacity of individual staff members, particularly during peak usage hours.

Santos and Bautista (2020) documented the specific error patterns that emerged from this attentional overload in manual multi-service environments. The most common errors were timer resets that occurred when staff attention was diverted by a printing or photocopying request, resulting in undercharging for computer rental; print job count errors that occurred when multiple jobs were submitted near-simultaneously; and change-giving errors in cash transactions that occurred when staff were simultaneously managing other service requests. The researchers found that these error patterns were not random but systematic — they clustered predictably around peak usage periods and the specific moments when multiple service demands coincided. This finding suggests that the errors were not attributable to individual staff incompetence but to fundamental limitations of human attentional capacity under conditions of simultaneous multi-task demand.

Santos and Bautista (2020) concluded that a unified management system capable of handling multiple service types under a single interface was the most effective solution to this operational challenge. They specified that the key design requirement was not merely the automation of individual service management functions but their integration within a

single attention field — a unified display that allowed staff to monitor all active services simultaneously without switching between different tools, records, or mental frameworks. This requirement for integrated simultaneous visibility is precisely what the Q-Solutions monitoring dashboard is designed to provide, making this study a particularly direct precedent for the current research.

Reyes and Dela Cruz (2021) investigated the adoption of cashless and semi-automated payment systems in provincial Philippine business centers, with a focus on establishments in Visayas and Mindanao. Their findings revealed that even in areas with lower average digital literacy, customers responded positively to transparent, system-generated billing as long as the interface was simple and the output — typically a printed receipt or a screen display — was clear and easy to understand. The researchers identified a "digital trust threshold," noting that customers who had experienced even one instance of digital system-generated billing that they could verify were significantly more likely to trust subsequent transactions, compared to customers who had no prior exposure to such systems.

Reyes and Dela Cruz (2021) also examined the role of staff behavior in mediating the customer experience of digital billing systems, finding that the manner in which staff introduced and explained the system to first-time customers significantly influenced initial acceptance. Establishments that trained staff to briefly explain the timer and billing system to new customers experienced much smoother adoption compared to those where staff simply operated the system without explanation, leaving customers uncertain about what was happening. This finding underscores the importance of staff training and communication protocols as complements to technical system deployment — a consideration that informed the staff training component of the Q-Solutions system implementation plan.

Aquino (2022) studied the operational practices of business centers located within commercial malls and supermarkets in Central Visayas, a geographic area of direct relevance to the Q-Solutions context. His research found that business centers in high-footfall commercial environments faced unique operational challenges not present in standalone business centers: higher transaction volumes due to the general foot traffic of the commercial environment, greater

diversity in customer demographics including both regular patrons and incidental visitors, increased scrutiny from mall or supermart management regarding professional conduct and receipt issuance compliance, and the expectation by customers of service standards comparable to other commercial establishments in the same environment.

Aquino (2022) found that business centers in commercial environments that operated with manual management systems were at a particular disadvantage relative to their counterparts in standalone locations, because the higher transaction volumes of commercial environments amplified the frequency of manual system failures — errors, disputes, and delays — that might have been manageable at lower volumes. His research documented several cases in which business centers in commercial environments had faced formal complaints from mall or supermart management following customer dispute incidents that attracted negative attention from other shoppers, highlighting the reputational stakes of billing accuracy and transparency in these high-visibility settings. These findings directly inform the priority given to billing accuracy and dispute prevention in the design of the Q-Solutions system.

Aquino (2022) also examined the specific service mix common to business centers in Central Visayas commercial environments, confirming that the most common combination was computer rental, document printing, and photocopying, precisely the services offered by Q-Solutions. He noted that this service combination created a particular billing challenge because each service type follows a different pricing logic: computer rental is time-based, printing is volume-based, and photocopying can be priced per page or per job. A management system that can accommodate all three billing logics within a unified interface was identified as a significant operational advantage, eliminating the need for staff to manage different billing methods simultaneously. Aquino's identification of this specific configuration as both common and challenging in the Central Visayas context makes his study the most geographically and operationally specific precedent for the Q-Solutions system design.

Lim and Fernandez (2023) examined student-patronized business centers near educational institutions in Cebu City, finding that the student demographic was particularly sensitive to billing transparency and particularly likely to patronize establishments that offered verifiable, system-generated billing. Their study found that 74% of student respondents stated that they would prefer a business center with a digital timer and automated receipt over one with manual logging, even if the prices were identical. This preference was most strongly expressed among college students, who cited familiarity with digital billing standards from e-commerce and online services and a general expectation that legitimate business establishments would use comparable standards in their physical operations.

Lim and Fernandez (2023) found that the student demographic also demonstrated high price sensitivity combined with relatively low sensitivity to minor service quality variations, making billing accuracy and transparency — which directly affect the perceived value of the service — disproportionately important drivers of patronage decisions compared to the experience in more affluent customer segments. Students who felt they had been overcharged, or even who suspected they might have been overcharged without the means to verify the bill, were highly likely to share their negative experience with peers through social networks, amplifying the reputational consequences of billing disputes beyond their direct impact on the individual transaction. This finding highlights the particular importance of billing transparency for business centers serving student populations, a significant component of Q-Solutions' customer base given the proximity of educational institutions to its supermart location.

Mendoza (2020) explored the relationship between system transparency and brand reputation in Philippine service SMEs, finding that businesses perceived as operating with transparent, accountable systems were significantly more likely to receive positive word-of-mouth referrals. In a market environment where small businesses compete heavily on trust and personal recommendation, the reputational benefits of transparent billing technology represent a meaningful competitive advantage that extends beyond the immediate operational context. Mendoza's research found that customer willingness to recommend a business to peers increased by 34% when the business used digital, verifiable billing compared to manual methods, and that this recommendation effect was strongest in community-oriented commercial settings — precisely the type of environment in which Q-Solutions operates.

Mendoza (2020) also examined the negative reputational impact of billing disputes, finding that customers who had experienced or witnessed a billing dispute at a particular establishment were significantly less likely to return and significantly more likely to actively discourage others from patronizing the business. The asymmetry between the positive impact of transparent billing and the negative impact of billing disputes — with the negative impact being disproportionately large relative to the positive — implies that the primary strategic imperative for billing system design is not merely to create positive impressions but to avoid the disproportionate reputational damage associated with perceived billing unfairness.

This finding reinforces the priority of dispute prevention as a design criterion for the Q-Solutions monitoring system.

Cruz and Villanueva (2022) investigated the specific challenges of cash management and daily reconciliation in Philippine SME service businesses, finding that the reconciliation of cash receipts with transaction records at the end of each business day was a major source of operational stress and was associated with a high incidence of discrepancies in manual-system environments. Their study found that average daily reconciliation time in manual-system businesses was 45 minutes per staff shift, compared to an average of 12 minutes in businesses using automated transaction logging systems. More significantly, the researchers found that residual discrepancies — cash amounts that could not be reconciled with available records — were present in 67% of manual-system businesses on any given day, compared to 8% in automated-system businesses.

Cruz and Villanueva (2022) analyzed the organizational impact of persistent reconciliation discrepancies in manual-system businesses, finding that they created a chronic atmosphere of mutual suspicion between owners and staff, as unresolved discrepancies were often interpreted as evidence of staff dishonesty in the absence of any alternative explanation. This atmosphere of suspicion had measurable negative effects on staff morale, staff retention, and the quality of owner-staff communication, creating a destructive cycle in which the operational failures of the manual system generated organizational dysfunction that further degraded operational

performance. The introduction of automated transaction logging eliminated the ambiguity that sustained this cycle by providing a complete, objective record against which cash receipts could be verified, thereby resolving discrepancies through factual investigation rather than interpersonal suspicion.

Soriano and Pascual (2021) examined the accessibility of business management technology for SME owners in provincial Philippine settings, finding that the primary barriers to adoption were perceived cost, perceived complexity, and the absence of locally available technical support. However, their study found that these barriers were most often based on outdated or inaccurate perceptions, as the actual cost of locally developed or adapted systems had decreased significantly in the years preceding the study, and the complexity of modern user interfaces had been substantially reduced through advances in user experience design. The researchers identified a significant gap between SME owners' perceptions of digital system accessibility and the actual current landscape of available options, suggesting that adoption rates were being suppressed partly by information gaps that could be addressed through targeted awareness and education initiatives.

Soriano and Pascual (2021) also found that demonstration-based introduction to digital management systems — where potential adopters could see and interact with the system in operation in a comparable business setting — was significantly more effective in overcoming adoption resistance than brochure or presentation-based information provision. This finding suggests that peer examples and live demonstrations play a critical role in the diffusion of business management technology in Philippine SME contexts, and that the documented adoption of the Q-Solutions system may itself serve a demonstration function for other business center operators in the same region considering similar transitions.

The collective findings of these local studies present a detailed and contextually specific picture of the challenges and opportunities facing Philippine business centers in their management of service operations. The recurring themes of billing dispute prevention, regulatory compliance, multi-service operational complexity, student customer expectations, reputational stakes of transparency, reconciliation efficiency, and adoption barrier reduction collectively define the problem space within which the Q-Solutions system was developed. Each local study contributes a distinct dimension to this picture, and together they provide an empirical foundation that both validates the need for the system and informs the specific design priorities that shaped its development.

2.3 Related Literature

The theoretical and technical foundations underpinning the Q-Solutions Business Center Monitoring System draw from a rich and diverse body of literature spanning the fields of system automation, database management, human-computer interaction, service operations management, business intelligence, and organizational behavior. This section presents a comprehensive review of the key conceptual frameworks and technical principles that inform the design, architecture, functionality, and implementation strategy of the proposed system. The literature reviewed here does not merely provide background context but actively shapes the design decisions documented in subsequent chapters, with specific theoretical frameworks providing direct guidance for specific system components and features.

The principles of System Automation suggest that replacing human-led repetitive tasks with software reduces the cognitive load on employees (Schildt, 2019). Cognitive load theory, originally developed in educational psychology by Sweller (1988) to explain the limitations of instructional design that overwhelmed learners' working memory capacity, has been productively applied in organizational and systems design contexts to explain why complex manual workflows generate systematic errors and inefficiencies. Working memory, the component of human cognition responsible for holding and manipulating information in real time, has a finite capacity typically estimated at the simultaneous management of five to nine distinct information units. Tasks requiring the tracking of multiple concurrent timers, each associated with different

usage rates and service types, quickly exhaust this capacity, leading to errors, omissions, and the degradation of attention to other important aspects of the work environment.

Schildt (2019) extended cognitive load theory to the context of business process automation, arguing that the strategic purpose of well-designed automated systems is not merely to increase speed or accuracy on individual tasks but to fundamentally reallocate cognitive resources within the work environment. By automating the routine monitoring and calculation tasks that consume the largest share of working memory in service environments, automated systems free cognitive capacity for the higher-order functions — customer relationship management, problem-solving, quality judgment, creative adaptation to novel situations — that remain beyond the capability of current automation technology and that constitute the most distinctive human contribution to service quality. This perspective reframes the value of automation not as the replacement of human judgment but as its liberation for higher-value applications.

In the context of business centers, the use of a Relational Database Management System (RDBMS) is essential for maintaining data integrity and ensuring that transaction history cannot be easily tampered with (Pratt and Last, 2020). The RDBMS represents the most widely adopted and thoroughly validated approach to structured data management in business information systems, offering a combination of data organization, query capability, integrity enforcement, and transaction management that is uniquely suited to the requirements of financial record-keeping. Pratt and Last's comprehensive treatment of RDBMS principles in applied business contexts provides a detailed technical foundation for understanding how relational database design supports the operational requirements of the Q-Solutions monitoring system.

Pratt and Last (2020) emphasized that the ACID properties of relational database transactions — Atomicity, ensuring that transactions are completed in full or not at all; Consistency, ensuring that all database constraints are maintained across every transaction; Isolation, ensuring that concurrent transactions do not interfere with each other; and Durability, ensuring that committed transactions persist even in the event of system failure — provide guarantees of data reliability that are particularly critical in financial transaction environments. In a business center context, ACID compliance ensures that a session record, billing calculation, and transaction log entry are

always created together or not at all, preventing the partial record states that were a major source of reconciliation difficulties in the pre-digital management environment.

Norman (2018), in his foundational work on user-centered design, argued that the design of interactive systems should be guided by a deep understanding of human cognitive and perceptual capabilities and limitations rather than by technical elegance or feature completeness. His concept of low-friction design holds that the most effective service management interfaces are those that minimize the number of cognitive decisions, physical actions, and time intervals required to complete routine operational tasks, reserving the user's attentional resources for the exceptional situations that genuinely require human judgment and discretion.

In a fast-paced service environment, every unnecessary step in a workflow represents an opportunity for error, delay, and customer frustration, and the cumulative impact of these unnecessary steps on operational efficiency can be substantial.

Norman (2018) identified several specific design principles that contribute to low-friction interfaces in service management contexts: affordance, ensuring that interface elements visually communicate their function and invite the appropriate action; feedback, ensuring that every user action produces an immediate, clear, and informative system response; constraints, limiting the range of possible actions at each decision point to those that are appropriate for the current context; and consistency, ensuring that similar functions are represented and operated in the same way throughout the system. These principles were directly applied in the design of the Q-Solutions monitoring system interface, with particular attention to the session initiation workflow, the billing calculation display, and the transaction completion process.

Codd (1970), whose pioneering work on the relational model of data management established the theoretical foundation for all subsequent RDBMS development, defined a set of mathematical principles for organizing data into structured relations that support powerful, flexible, and efficient data retrieval and manipulation. The enduring relevance of Codd's relational model seven decades after its publication is a testament to the fundamental correctness of its underlying logic and its

remarkable capacity to accommodate the data management requirements of an enormous diversity of application domains. In the context of the Q-Solutions system, the relational model enables the representation of complex real-world relationships — between customer sessions, service types, billing rates, staff accounts, transaction records, and summary reports — in a structured form that supports both the real-time operational queries required during service delivery and the analytical queries required for management reporting.

Sommerville (2016), in his authoritative treatment of software engineering principles and methodologies, outlined the importance of requirements elicitation as the foundational activity in any system development project. He argued that systems developed without deep and systematic understanding of the operational context and user workflows frequently fail to achieve adoption or to deliver the expected benefits, regardless of their technical sophistication, because they address a model of the work environment rather than the actual work environment. Sommerville's emphasis on user-centered requirements elicitation directly informed the research methodology employed in this study, which began with an extensive investigation of Q-Solutions' current manual workflows, staff pain points, customer experience issues, and management information needs before proceeding to system design and development.

Sommerville (2016) also stressed the importance of iterative development and user feedback in system refinement, arguing that no system design, regardless of how carefully it is developed from requirements elicitation, will perfectly match user needs on its first deployment. The process of use inevitably reveals requirements and usability issues that were not apparent in the pre-deployment investigation, and the ability to incorporate this feedback into system improvements is a critical determinant of long-term system success. This principle informed the iterative testing and refinement process employed during the development of the Q-Solutions system, with successive prototype evaluations by staff and management informing a series of interface and functionality improvements prior to final deployment.

Hammer and Champy (1993) introduced the concept of business process reengineering as a framework for thinking about the radical redesign of organizational workflows to take advantage of the capabilities of information technology. They argued that the most transformative benefits of technology adoption come not from using technology to automate existing manual workflows but from using the capabilities of technology to fundamentally reconceive the workflow itself. This principle is highly relevant to the Q-Solutions system development, which did not merely digitalize the existing manual process of timer tracking and manual billing but redesigned the entire service management workflow around the capabilities of automated timing, real-time monitoring, and integrated transaction logging.

Hammer and Champy (1993) emphasized that genuine process reengineering requires a willingness to question fundamental assumptions about how work should be organized, including assumptions so deeply embedded in current practice that they are rarely consciously articulated. In the context of the Q-Solutions billing process, a fundamental embedded assumption of the manual system was that staff members were responsible for tracking service time — a reasonable assumption given the absence of alternatives, but one that introduced all of the human error and customer trust problems documented in the local and foreign studies reviewed above. The Q-Solutions system reengineers this assumption by making the system, rather than the staff member, responsible for time tracking, fundamentally eliminating the error source rather than merely reducing its frequency.

Davenport (1998) extended the business process reengineering framework to address the role of enterprise resource planning and integrated information systems in organizational transformation. His research on large-scale system implementations identified several critical success factors that determined whether system deployments achieved their intended benefits, including senior management commitment, adequate staff training, realistic expectations about implementation timelines, and the alignment of system design with actual operational workflows rather than idealized versions of those workflows. While Davenport's research focused primarily on large enterprise contexts, subsequent researchers including Muscatello and Chen (2008) demonstrated the applicability of these success factors to SME contexts with appropriate scaling adjustments.

Turban, Sharda, and Delen (2014) provided a comprehensive framework for understanding how business intelligence and decision support systems generate value in organizational contexts at different scales. Their work on dashboard design is particularly relevant to the Q-Solutions monitoring system, as they outlined in detail the key principles of effective dashboard architecture from both a technical and cognitive perspective. They argued that an effective operational dashboard must accomplish three distinct functions simultaneously: it must provide an at-a-glance summary of current operational status that allows a manager to quickly assess whether the business is operating normally or whether intervention is required; it must provide sufficient detail at appropriate levels of granularity to support diagnostic investigation when anomalies are detected; and it must enable action — it must not merely inform but must be directly connected to the controls that allow issues to be addressed.

Turban et al. (2014) also addressed the temporal dimension of dashboard design, distinguishing between real-time dashboards that display current operational status, historical dashboards that display trends and patterns over time, and analytical dashboards that enable interactive exploration of data across multiple dimensions. They argued that operational environments like business centers require primarily real-time dashboards for day-to-day management but benefit significantly from historical and analytical views for strategic planning and performance evaluation. The Q-Solutions system incorporates both real-time monitoring functions, in the form of the active session dashboard, and historical reporting functions, in the form of daily, weekly, and monthly sales summaries, reflecting the dual operational and analytical value of comprehensive transaction data.

Pressman (2014), in his comprehensive treatment of software engineering methodology, discussed in detail the critical role of systematic testing in ensuring the quality and reliability of deployed software systems. He outlined a hierarchical testing strategy progressing from unit testing of individual software components, through integration testing of component interactions, to system testing of overall functional and performance requirements, and finally to user acceptance testing in conditions approximating the real deployment environment. Pressman argued that each level of testing serves a distinct and irreplaceable function in ensuring system quality: unit tests verify the correctness of individual components in isolation; integration tests verify that components interact correctly when combined; system tests verify that the assembled

system meets its specified requirements; and acceptance tests verify that the system meets the actual needs of real users in real operational conditions.

Laudon and Laudon (2018) examined the role of management information systems in improving the quality of decision-making at all levels of an organization, from frontline operational decisions made by service staff to strategic decisions made by business owners and senior managers. They proposed a framework for evaluating MIS quality based on three dimensions: the quality of the information produced, defined by its accuracy, completeness, timeliness, and relevance; the quality of the system through which the information is delivered, defined by its reliability, ease of use, and flexibility; and the quality of the organizational context in which the system is deployed, defined by the degree to which the system is integrated with and supported by organizational processes, skills, and culture. The Q-Solutions system was evaluated against all three dimensions during its design and development, with particular attention to the information quality dimension, given the central importance of accurate, timely billing and transaction data to the system's core purposes.

Laudon and Laudon (2018) also provided a useful framework for thinking about the different types of decision support provided by management information systems at different organizational levels. At the operational level, the Q-Solutions system supports real-time service delivery decisions: which computer to assign to an incoming customer, how to respond to a printer malfunction, when to prompt a customer about session extension.

At the managerial level, it supports short-term resource allocation and scheduling decisions based on usage pattern data. At the strategic level, it supports longer-term business development decisions — whether to add service capacity, how to adjust pricing, whether to introduce new service types — based on historical performance data and trend analysis.

O'Brien and Marakas (2011) explored the concept of system integration as a source of distinctive organizational value in business information system contexts. They argued that the greatest operational benefits of information technology adoption are achieved not through the deployment of individual software tools, each addressing a specific operational function in isolation, but through the integration of these tools into coherent information systems that share data seamlessly across functional boundaries. The value of integration derives from the

elimination of the informational seams between functional areas — the points at which data must be manually transferred, re-entered, or reconciled across system boundaries — which are both major sources of operational error and significant consumers of staff time and attention.

O'Brien and Marakas (2011) provided a taxonomy of system integration types, ranging from simple data exchange between independent systems at the lowest level of integration to fully unified platforms in which all functional modules share a common data repository at the highest level. They argued that the level of integration appropriate for a given organizational context depends on the degree of informational interdependence among the functions being supported: where functions are closely coupled — where the output of one function is a direct input to another, as is the case with session timing and billing calculation in a business center — the highest level of integration, through a unified data repository and integrated functional modules, is both necessary and most valuable. This principle directly informed the architectural decision to develop the Q-Solutions system as a fully integrated platform rather than a collection of loosely connected tools.

Patterson and Hennessy (2014) addressed the hardware and system architecture dimensions of reliable computing, discussing the principles of redundancy, fault tolerance, and graceful degradation that distinguish robust computing systems from fragile ones. In a business environment where system downtime directly translates to lost revenue and customer dissatisfaction, the reliability of the monitoring system is as important as its functional capability. A system that provides accurate billing and excellent reporting but fails unpredictably or loses data during power interruptions provides only limited value compared to a system that combines functional capability with architectural reliability.

Fitzsimmons and Fitzsimmons (2011) examined service operations management from an integrated perspective that considered the design of service delivery systems as a whole, including the physical environment, the human service providers, the supporting technology infrastructure, and the customers themselves. Their concept of the "service package" — the

bundle of tangible and intangible elements that constitute the customer's service experience — provides a useful framework for understanding how the Q-Solutions monitoring system contributes to service quality beyond its direct operational functions. By enabling accurate billing, transparent timing, and professional receipt issuance, the system contributes to the professionalism and reliability dimensions of the service package, which Fitzsimmons and Fitzsimmons identified as among the most important determinants of customer satisfaction in business service contexts.

McAfee and Brynjolfsson (2012) examined the relationship between information technology investment and business performance at the organizational level, finding that the performance differential between IT-intensive and IT-lagging businesses in the same industry had been growing steadily in the decade preceding their study. They argued that this growing performance gap was attributable not merely to the direct operational benefits of specific technologies but to the compounding advantages that accrue to organizations that develop the institutional capabilities — data literacy, process discipline, analytical culture — associated with effective technology use. Their findings suggest that the value of the Q-Solutions system extends beyond its immediate operational impact to encompass the development of an organizational capability for data-driven management that will generate compounding returns as the business matures and the system's data accumulates.

Turban and Volonino (2015) examined the security and access control dimensions of business information systems, arguing that well-designed access control systems serve a dual function: they protect sensitive business data from unauthorized external access, and they provide an internal accountability framework that deters and detects unauthorized internal actions. In the context of the Q-Solutions system, access control is implemented through user authentication and role-based permissions, ensuring that each system action is traceable to a specific authenticated user and that access to sensitive functions such as transaction modification and report generation is appropriately restricted. This access control framework directly addresses the internal fraud risk documented by Patel and Okonkwo (2022) and Cruz and Villanueva (2022) in relation to manual billing systems.

Collectively, the related literature provides a rich, multi-dimensional theoretical and technical framework that transforms the empirical evidence of the foreign and local studies into actionable design principles. The convergence of insights from cognitive psychology, database theory, software engineering, human-computer interaction, service operations management, business intelligence, organizational behavior, and system security underscores the genuinely multidisciplinary nature of effective automated management system design and validates the comprehensive, principled approach taken in the development of the Q-Solutions Business Center Monitoring System.

2.4 Synthesis of the Reviewed Literature

The comprehensive review of foreign studies, local studies, and related literature presented in the preceding sections of this chapter constitutes one of the most extensive and contextually specific bodies of evidence available in support of the development of an automated monitoring and billing system for a multi-service business center in the Philippine setting. This synthesis draws together the key themes, converging findings, contrasting perspectives, and identified research gaps from across the entire body of reviewed material, articulating the collective intellectual foundation upon which this study stands and clearly delineating the specific contribution that the Q-Solutions Business Center Monitoring System makes to both theoretical knowledge and practical business management.

The reviewed literature and studies collectively and unambiguously underscore a global and local shift from manual to automated monitoring systems, driven by the documented benefits of automation in reducing human error, improving revenue accuracy, enhancing customer trust, strengthening regulatory compliance, and enabling data-driven management decision-making. This shift is documented across a remarkable diversity of geographic contexts — from the United States and United Kingdom to South Korea, Vietnam, India, Nigeria, Australia, Canada, Singapore, and across multiple regions of the Philippines — and across a range of business types and scales, from large multi-branch franchise operations to individual small business owners operating from single

locations. The consistency of this trend across such varied contexts speaks to the universality of the underlying principles: human cognitive limitations, the trust-building function of transparent and verifiable records, and the organizational value of real-time data access are not culturally or contextually specific phenomena but fundamental features of human cognition and organizational dynamics that manifest predictably across the full spectrum of service business environments.

The quantitative evidence accumulated across the foreign studies reviewed in Section 2.1 is particularly striking in its consistency and magnitude. Revenue accuracy improvements ranging from 12% to 30%, reductions in customer complaints of up to 58%, decreases in staff training time of approximately 35%, improvements in customer satisfaction scores averaging 18 percentage points, reductions in workplace stress scores of up to 28%, and improvements in customer recommendation rates of up to 34% collectively represent a compelling empirical case for system adoption. The fact that these figures derive from studies conducted using different research methodologies — longitudinal tracking studies, quasi-experimental designs, cross-sectional surveys, case studies, and mixed-methods investigations — in different countries and business contexts significantly strengthens their credibility. Findings that replicate across methodological approaches and contextual variations are considerably more robust than those that emerge from a single study type or single geographic context.

The local Philippine studies reviewed in Section 2.2 add essential contextual specificity to this global picture, documenting the particular ways in which the challenges of manual service management manifest in the Philippine SME environment. The prominence of billing disputes as a driver of customer dissatisfaction and staff stress (Villanueva and Mercado, 2022), the growing regulatory pressure from BIR and local government units that makes comprehensive transaction recording a compliance necessity (Garcia, 2021), the specific operational complexity of multi-service environments combining computer rental with printing and photocopying (Santos and Bautista, 2020; Aquino, 2022), the particular expectations of the student customer demographic (Lim and Fernandez, 2023), the reputational stakes of billing transparency in community commercial settings (Mendoza, 2020), and the cash management and reconciliation challenges of manual-system environments (Cruz and Villanueva, 2022) collectively define a

problem landscape that is both substantively important and insufficiently addressed by existing systems and research.

The related literature reviewed in Section 2.3 provides the theoretical and technical frameworks that transform this body of empirical evidence into coherent design principles. Schildt's (2019) cognitive load framework explains the mechanism through which automation improves staff performance and reduces error frequency. Pratt and Last's (2020) RDBMS principles explain the technical foundation for transaction data integrity and tamper resistance. Norman's (2018) low-friction interface design principles define the user experience requirements that must be met for a system to achieve genuine adoption in a fast-paced service environment. Codd's (1970) relational model provides the mathematical foundation for the database architecture that organizes and relates the diverse data generated by multi-service business center operations.

Turban et al.'s (2014) dashboard design principles define how real-time operational monitoring information should be structured and presented to maximize its practical utility. Hammer and Champy's (1993) process reengineering framework provides the conceptual basis for designing the entire service management workflow around the capabilities of automation rather than merely digitizing the existing manual workflow.

A critical theme that emerges from the synthesis of all three bodies of reviewed material is the importance of system design fit — the degree to which a system's design matches the specific operational context, user characteristics, and organizational requirements of the environment in which it is deployed. This theme is most explicitly articulated by Nguyen (2021) and Sommerville (2016), but it is implicit in the findings of numerous other reviewed works: systems that fit their operational context achieve adoption and deliver benefits; systems that do not fit their context fail to achieve adoption regardless of their technical sophistication. The Q-Solutions system was designed from the beginning with explicit attention to contextual fit, grounded in the findings of the local studies reviewed above and refined through the requirements elicitation process described in Chapter 3. Every design decision — the specific service types accommodated, the billing logic implemented, the interface language used, the receipt format generated, the reporting periods supported — reflects a deliberate effort to match the system's capabilities to the actual operational reality of Q-Solutions Business Center.

While existing studies have focused heavily on large-scale enterprises, basic internet cafes limited to computer rental services, or general SME digitalization without addressing the specific operational challenges of multi-service business centers, there is a demonstrated and significant gap in the research and systems design literature specifically addressing establishments that offer the combination of computer rental, document printing, and photocopying services within a single unified operational framework. This gap is not merely academic but has practical consequences: business center operators in the Philippines currently lack access to purpose-built management systems that address their specific service mix, and are instead compelled to choose between general-purpose POS systems that do not accommodate time-based billing for computer rental, basic timer software that does not accommodate per-page billing for printing and photocopying, or expensive enterprise platforms designed for much larger operations with different operational characteristics.

This project directly and specifically addresses this gap by creating a unified system that manages all three service types under a single automated dashboard designed for the particular operational context of a Philippine business center in a high-traffic commercial environment.

The system's architecture synthesizes the RDBMS principles of Pratt and Last (2020), the interface design philosophy of Norman (2018), the dashboard design principles of Turban et al. (2014), the process reengineering framework of Hammer and

Champy (1993), and the system integration principles of O'Brien and Marakas (2011) into a coherent design that is simultaneously technically rigorous and practically accessible to the staff and management of a small business center operating in a supermart branch environment.

The synthesis of reviewed literature also reveals a second-order contribution of automated monitoring systems that transcends the immediate operational improvements in billing accuracy, customer satisfaction, and reconciliation efficiency. The data generated by a comprehensive monitoring system — detailed, accurate, complete, and continuous records of all transactions, sessions, and service usage — represents a business intelligence asset of substantial and growing value over time. As transaction records accumulate, they enable increasingly sophisticated analyses of business performance: identification of revenue trends, detection of underperforming service types, optimization of pricing structures, evidence-based assessment of the impact of

operational changes, and preparation of comprehensive, audit-ready financial records. This transformation of operational data into strategic intelligence represents a qualitative enhancement of the business owner's managerial capability that compounds in value as the system matures and the data archive deepens, a benefit documented empirically by McAfee and Brynjolfsson (2012) and Anderson and Reyes (2019) and theorized by Laudon and Laudon (2018) and Turban et al. (2014).

The human dimension of system impact, consistently documented across the reviewed foreign studies (Williams and Thompson, 2021; Martinez and Liu, 2023) and theoretically grounded in Schildt's (2019) cognitive load framework, represents an additional dimension of system value that deserves explicit recognition in any comprehensive synthesis of the literature. The reduction of staff stress, the elimination of the socially uncomfortable dynamic of defending manual billing records against customer challenges, the improvement of staff confidence through the availability of authoritative system records, and the reallocation of cognitive capacity toward higher-value customer service functions collectively constitute a meaningful improvement in the quality of the work environment for business center staff. In an operational context characterized by high transaction volumes, time pressure, and close customer contact, this improvement in staff working conditions is likely to translate into measurable improvements in service quality, staff retention, and the overall management health of the business.

The regulatory compliance dimension of automated system adoption, highlighted by Garcia (2021), Patel and Okonkwo (2022), and Cruz and Villanueva (2022), represents an increasingly important driver of digitalization in the Philippine SME context as regulatory requirements become more stringent and their enforcement more consistent. The ability of the Q-Solutions system to generate complete, timestamped, unalterable transaction records on demand positions the business not merely to comply with current requirements but to respond effectively to any future intensification of regulatory scrutiny, providing a degree of compliance resilience that manual systems cannot offer.

In conclusion, the synthesis of reviewed literature establishes beyond reasonable doubt that the Q-Solutions Business Center Monitoring System is a necessary, evidence-based, and theoretically well-grounded response to a real, documented, and previously inadequately addressed set of operational challenges faced by multi-service business centers in the Philippine commercial environment. The convergence of international empirical research, Philippine-specific local studies, and foundational theoretical literature across multiple academic disciplines creates a comprehensive, multi-dimensional justification for the system's development that is both academically rigorous and practically compelling. The identification of a specific and significant research and design gap — the absence of a purpose-built, unified monitoring system for the computer rental, printing, and photocopying service combination in Philippine SME business center contexts — positions the Q-Solutions system as a novel and meaningful contribution to both academic knowledge in the field of information systems for SME management and practical business operations for business center operators in the Central Visayas region and throughout the Philippines.

The system developed in this study does not merely replicate or adapt existing solutions but synthesizes established principles with original contextual design to address a real, documented, and previously unaddressed operational need. Its development is grounded in the most relevant and current available evidence from both international and Philippine sources, designed in accordance with the best available theoretical frameworks from multiple contributing disciplines, and validated against the actual operational requirements of its target environment. The review of related literature and studies presented in this chapter provides the intellectual foundation for every subsequent design decision, implementation choice, and evaluation criterion documented in the chapters that follow.

CHAPTER 3

RESEARCH METHODOLOGY

This chapter details the systematic approach used to design, execute, and validate the study. It presents the selected research framework, the software development life cycle, the techniques for data gathering, and the strategies for testing and evaluating the system.

3.1 Research Design

The study will use a **developmental research design**, focusing on the design, development, and evaluation of the **Q-Solutions Automated Management and Monitoring System**. Developmental research is defined as the systematic study of designing, producing, and assessing tools, processes, and software products to ensure they meet internal consistency and operational effectiveness. This specific design is chosen because the primary objective of this research is to solve a practical organizational problem by engineering a technical product. The research focuses on three key processes:

- **Design Phase:** Conceptualizing the user interfaces, structural data flow, and underlying system architectures tailored specifically to the operational context of Q-Solutions.
- **Development Phase:** Translating these architectural concepts into a working application using appropriate programming languages, algorithms, and logical frameworks.
- **Evaluation Phase:** Utilizing empirical validation methods to determine if the finalized prototype satisfies performance benchmarks, functions reliably, and addresses the administrative issues of the organization.

3.2 System Development Methodology

To execute the technical requirements of this study, the **Agile Development Model** will be used. The Agile framework operates on iterative and incremental development cycles, which allow the system to evolve dynamically based on ongoing feedback and technical evaluations. This model prevents project rigidity, ensuring that changing software requirements can be handled effectively throughout the system lifecycle.

3.2.1 Requirements Analysis

This initial phase aims to establish a clear baseline of what the software must accomplish. The primary focus is on gathering data directly from Q-Solutions staff.

- **Data Collection:** The researchers will conduct semi-structured interviews and workplace observations to understand the current bottlenecks in daily operations.
- **Operational Scope:** Information regarding the manual tracking of tasks, shifting schedules, worker performance metrics, and current billing workflows will be fully documented.
- **Outcome:** A comprehensive Software Requirements Specification (SRS) document that outlines the definitive functional and non-functional requirements of the proposed system.

3.2.2 System Design

The structural blueprint of the application is constructed during this phase based on the compiled requirements.

- **User Interface (UI) Design:** Wireframes and interactive user interface mockups will be created for the administration and staff dashboards. The layouts will emphasize simplicity and ease of navigation.
- **Database Schema:** An Entity-Relationship Diagram (ERD) will be formulated to map out tables, attributes, primary/foreign keys, and standard data constraints. This ensures a structured and organized database layout that eliminates data redundancy.

- **Architecture:** System data flows and permission layers will be established to control access rights between general staff and system administrators.

3.2.3 Development

This phase involves the actual programming and engineering where design models are converted into source code.

- **Timer Algorithms:** Complex programmatic logic and specialized algorithms will be coded to log operational runtimes, automated inactivity timeouts, and accurate timestamps.
- **Billing Modules:** Code structures will be built to automatically convert logged times into monetary totals, apply varying billing rates, and generate formatted digital invoices.
- **Technology Stack:** The development environment will leverage appropriate frontend, backend, and relational database technologies to ensure high performance and data security.

3.2.4 Testing

The completed system components will undergo strict quality assurance steps to uncover software bugs and functional anomalies.

- **Unit and Integration Testing:** Modules like time-tracking and billing will be tested both independently and in conjunction with each other to evaluate performance stability.
- **Data Accuracy Verification:** The calculation engines will be subjected to synthetic test cases to verify that time logs and billing balances contain zero mathematical errors.
- **Refinement:** Code revisions and logic optimizations will be applied directly when any operational errors or discrepancies are discovered.

3.2.5 Deployment (Prototype)

The final stage establishes the working framework in a live workspace environment.

- **Controlled Installation:** The system prototype will be deployed onto selected hardware within a designated section of the Q-Solutions office.
- **Staff Use:** Staff members will utilize the system during daily operations to manage real assignments under direct observation.
- **Feedback Gathering:** User experiences and system logs gathered from this deployment will serve as empirical data for the final evaluation phase, driving the next iteration of the software.

3.3 System Architecture

The **Q-Solutions Automated Management and Monitoring System** utilizes a structured **3-Tier Client-Server Architecture**. This architectural framework separates the application into three distinct logical layers: the presentation layer, the application logic layer, and the data management layer. This approach isolates specific functions to ensure optimal performance, simplified system maintenance, and enhanced data security across all connected computer units.

3.3.1 Presentation Layer

The presentation layer handles user interaction and the overall visual interface of the application.

- **User Interaction:** Operations staff and administrators interact with the system via a web-based dashboard or desktop frontend wrapper.
- **Client Capabilities:** The interface allows administrators to view real-time workstation statuses, manually adjust PC allocations, lock or unlock client terminals, and generate reports.
- **Decoupled Design:** This tier relies strictly on the underlying application layer to execute operations, meaning no raw business rules or direct database access scripts are run locally on the client interface.

3.3.2 Application Layer

The application layer acts as the primary engine of the system, handling the core business rules and workflows.

- **Process Coordination:** Sitting between the user interface and the database, this tier processes all incoming client commands.
- **Real-Time Calculations:** The server maintains independent background processes responsible for tracking individual PC session timers and processing complex billing logic.
- **Session Management:** It constantly calculates runtime balances based on user activity and automatically pushes terminal-lock commands when time limits or credits run out.

3.3.3 Data Layer

The data layer serves as the secure, centralized repository for all persistent system records.

- **Information Assets:** This tier handles the physical storage of critical operational records, including system event history, employee login logs, structural billing tariffs, and client session archives.
- **Data Integrity:** Managed by a relational database system, it handles high-volume transaction loads from the application layer while enforcing strict constraints to prevent data corruption or log entry collisions.

3.4 Tools and Technologies Used

To successfully implement the proposed 3-tier client-server architecture, a specific combination of frontend development, backend processing, database management, and utility tooling has been selected. The selection of these technologies is guided by principles of operational reliability, modularity, security, and compatibility with localized client-server deployment environments.

3.4.1 Frontend Technologies

- **HTML5 (HyperText Markup Language)**: Used as the foundation of the administrative web dashboard to establish semantic structure and properly organize layout components.
- **CSS3 (Cascading Style Sheets)**: Handles presentation aesthetics, styling rules, and layout structures to ensure the user interface appears clean, cohesive, and uniform.
- **JavaScript (ES6+)**: Handles dynamic behavior within the browser environment, powering client-side calculations, managing user actions, and running asynchronous background requests to communicate with the server without requiring full page reloads.
- **React / Bootstrap Frameworks**: Used to speed up frontend development. **Bootstrap** provides responsive, grid layouts and predefined UI styles, while **React** builds a modular, component-driven dashboard capable of efficiently rendering real-time updates as workstation statuses change.

3.4.2 Backend Technologies

- **PHP (Hypertext Preprocessor)**: Serves as the primary server-side programming language. It is responsible for intercepting incoming HTTP payload requests from client machines, verifying credentials, running billing rules, and executing queries against the database engine. PHP handles session tracking and provides a bridge between client requests and data states.

3.4.3 Database Management System

- **PostgreSQL (postgres.sql):** Selected as the primary relational database management system (RDBMS) for this study. PostgreSQL offers robust support for concurrent transactions, strict data type validations, and high reliability under continuous write loads. It securely processes financial records and continuous timestamp logs while preventing data loss or structural errors.

3.4.4 Development Tools

- **Visual Studio Code (VS Code):** Utilized as the primary integrated development environment (IDE). Its code compilation checking, built-in version control integrations, and terminal access streamline debugging across the frontend and backend tech stacks.
- **Laragon:** Used as the localized development environment server suite. Laragon provides an isolated, fast, and highly efficient stack for running the Apache/Nginx web

server and managing the PHP runtime. Its fast database management integrations and automatic local virtual hosts offer an optimal and stable environment for building and testing the system prototype before production.

3.5 Data Gathering Techniques

To establish a comprehensive baseline of requirements for the Q-Solutions Automated Management and Monitoring System, the researchers will apply triangulation through three primary data gathering techniques. This ensures the technical features align perfectly with active field realities.

3.5.1 Interviews

The researchers will conduct formal, semi-structured interviews with the branch supervisors of Q-Solutions.

- **Target Insights:** Discussions will target active operational pain points, recurring scheduling bottlenecks, current constraints in manual time tracking, and authorized administrative overrides.

- Operational Rules: These direct consultations will clarify administrative hierarchies and distinct permission controls necessary for designing the user clearance profiles.

3.5.2 Observations

Direct workplace observations will be executed on-site at the Q-Solutions branch located inside Iloilo Supermart.

- Target Insights: This process involves documenting exact employee workflows, customer service pipelines, queue response rates, and system loads during peak operational hours.
- Operational Rules: The data gathered will define user interface usability priorities and stress points, ensuring the software remains highly responsive and easy to navigate under busy conditions.

3.5.3 Document Analysis

The researchers will conduct a comprehensive review of the active manual logbooks, spreadsheets, and paper receipts currently used by the branch.

- Target Insights: This analytical review aims to extract structural data fields, tracking templates, and auditing parameters.
- Operational Rules: Identifying these legacy variables directly guides the engineering of the PostgreSQL database schema, ensuring all necessary financial data points, timestamp formats, and customer logs are retained.

3.6 Testing Procedures

To guarantee system stability, mathematical precision, and client satisfaction, the software prototype will undergo a rigorous multi-tiered verification framework.

3.6.1 Unit Testing

Unit testing isolates individual software functions to verify their logical correctness independently of external system modules.

- Focus Area: Primary focus will be dedicated to testing the underlying start, pause, resume, and stop timer algorithms.
- Execution: The researchers will execute controlled code assertions to ensure that millisecond conversions, state changes, and localized system timeouts respond accurately without performance delays.

3.6.2 System Testing

System testing verifies the operational harmony, database connectivity, and data pipelines between interconnected application components.

- Focus Area: Testing will focus on verifying that the automated billing module dynamically draws and processes active timestamp figures sent directly from the running session timers.
- Execution: The trial will ensure that data objects transfer across the 3-tier client-server architecture without dropping values or experiencing network latency errors.

3.6.3 User Acceptance Testing (UAT)

User Acceptance Testing involves deploying the functional system prototype directly to the operations team for hands-on evaluation. [\[1\]](#)

- Focus Area: Q-Solutions staff will evaluate the system's runtime ease of use, screen layouts, and workflow intuitive design during standard operational shifts.

- Execution: Feedback gathered during this phase will serve as empirical validation, determining whether the user interface is optimized for rapid use or requires final refinement.

3.7 Evaluation Criteria

The overall performance, corporate viability, and technical success of the engineered prototype will be measured against five distinct criteria:

- Functionality: Assessment of whether the system completely automates the legacy manual processes. It must successfully manage concurrent workstation instances and handle routine monitoring requests without requiring manual worker intervention.
- Usability: Evaluation of the interface layout and navigation speed. The frontend dashboard must prove simple and intuitive enough for staff to operate flawlessly during high-traffic shifts at the Iloilo Supermart branch.
- Accuracy: Strict mathematical verification of the billing engines. The automated billing outputs must perfectly match defined company rates, demonstrating zero calculation discrepancies across varying operational runtimes.
- Security: Verification of data protection layers within the system. The application must guarantee that financial logs, transaction records, and administrative settings are protected from unauthorized user access or malicious data modifications.
- Efficiency: Measurement of time optimization metrics. The system must reduce the time required to generate comprehensive end-of-day operational and financial balance reports compared to manual logs.

3.8 Supporting Diagrams

Data Flow Diagram (DFD – Level 0)

The Level 0 Data Flow Diagram maps out the fundamental boundaries of the system, tracking how transactional, computational, and financial information moves between users, processing modules, and persistent database storage.

DFD Operational Processes:

1. **Staff/Admin Inputs:** The staff member logs into the presentation layer, selects the appropriate customer service profile, and initiates the runtime timer instance.
2. **System Calculations:** The central server monitors the elapsed time session. Upon completion, the application layer queries the defined pricing structures, computes the precise financial balance, and prompts the payment window.
3. **Database Storage:** The finalized transaction details, historical timestamps, audit numbers, and associated financial logs are securely committed to the centralized PostgreSQL data store for future report generation.

Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) serves as the structural blueprint for the PostgreSQL database. It models the organization's information assets by defining explicit logical relationships, primary and foreign keys, PostgreSQL-specific data types, and operational integrity constraints.

Data Dictionary and Entity Specifications

To ensure database consistency and optimization, the data types, descriptions, and key structural properties of each entity are mapped out below:

1. Staff Entity

This entity tracks the administrative and operational users authorized to navigate the system interface.

- StaffID (*Integer, Primary Key*): A unique auto-incremented identifier for each user profile.
- Name (*Varchar, 100*): The full legal name of the employee.
- Username (*Varchar, 50*): Unique login credential used for authentication.
- Password (*Varchar, 255*): Secured login token, encrypted on the server layer using PHP `password_hash()`.
- Role (*Varchar, 20*): User access authorization layer (e.g., 'Administrator', 'Staff').

2. PC_Unit Entity

This entity tracks the physical workstations being managed and monitored inside the branch.

- UnitID (*Integer, Primary Key*): A unique identifier assigned to each computer terminal.
- UnitNumber (*Integer*): The display label of the computer unit (e.g., 1, 2, 3).
- Status (*Varchar, 20*): The live status condition of the workstation (e.g., 'Available', 'Occupied', 'Maintenance').

3. Service Entity

This entity stores the system's structural pricing definitions, detailing the specific operational types and billing scales.

- ServiceID (*Integer, Primary Key*): Unique identifier for each operational service option.
- ServiceName (*Varchar, 50*): The descriptive name of the service type (e.g., 'PC Rental', 'Black & White Printing', 'Colored Printing').
- RatePerHour (*Numeric, 8, 2*): The specific hourly monetary rate applied if the selection relies on runtime tracking.
- RatePerPage (*Numeric, 8, 2*): The static per-page cost tier applied for print job processes.

4. Transaction Entity

The central intersection entity of the schema, tracking every active tracking session, billing computation, and audited activity.

- TransID (*BigInteger, Primary Key*): A unique audit trace number generated upon record initiation.
- StaffID (*Integer, Foreign Key*): References the specific employee who initiated or cleared the customer request.
- UnitID (*Integer, Foreign Key*): References the exact computer workstation used during the session.
- ServiceID (*Integer, Foreign Key*): References the configuration rules applied for cost calculation.
- StartTime (*Timestamp*): The exact automated date and time when the session timer was activated.
- EndTime (*Timestamp, Nullable*): The automated log captured upon terminal completion.
- TotalAmount (*Numeric, 10, 2*): The finalized monetary sum calculated by the server's billing engine.

System Cardinality Rules and Normalization

The data schema follows Third Normal Form (3NF) parameters to eliminate data redundancies and enforce absolute integrity:

- Staff to Transaction (1 to Many): One staff member can manage and record zero to many (1 : 0..*) separate transactions over their operational work shift. However, each unique transaction record belongs strictly to the specific employee logged in when it occurred.
- PC_Unit to Transaction (1 to Many): A specific computer terminal can be used across zero to many (1 : 0..*) independent customer transactions over time. Conversely, a single transaction record maps back to exactly one physical computer unit session.

- Service to Transaction (1 to Many): A pricing model rule can apply to zero to many (1 : 0..*) separate customer transactions. Each completed transaction is calculated based on exactly one selected service type profile to prevent billing configuration conflicts.

Technical Relationship Specifications and Constraints

The business rules of Q-Solutions are enforced within the database schema using relational constraints. These dependencies define how data links across the tables to ensure complete data integrity.

1. Staff manages Transaction (1 to Many Relationship)

This link binds human accountability to the system's financial records.

- Database Mapping: The **StaffID** primary key in the Staff table maps to the **StaffID** foreign key in the Transaction table.
- Cardinality Logic (1:\text{N}): A single employee can launch, monitor, and clear zero to many (0..*) distinct transactions during their shift. However, any given transaction record is locked to exactly one (1) unique staff member account.
- Referential Integrity: An **ON DELETE RESTRICT** constraint is placed on this link. An employee profile cannot be deleted from the database if they are tied to existing historical transactions, preserving the audit trail for end-of-day reports.

2. Transaction involves PC_Unit (Many to 1 Relationship)

This linkage monitors asset utilization and prevents overlapping device allocations.

- Database Mapping: The **UnitID** primary key from the PC_Unit table maps to the **UnitID** foreign key in the Transaction table.
- Cardinality Logic (\text{N}:1): Over its lifecycle, a single physical computer terminal can be checked into and out of zero to many (0..*) separate customer transactions. Conversely, an individual transaction record can only track the runtime usage of exactly one (1) hardware unit at a time.

- Referential Integrity: The system uses a validation lock. When a transaction is initialized, the system switches the status field of the involved **PC_Unit** record from 'Available' to 'Occupied'. This prevents other server requests from assigning the same terminal until the current transaction records an **EndTime**.

3. Transaction includes Service (Many to 1 Relationship)

This relationship provides the automated billing engine with its calculations and pricing rules.

- Database Mapping: The **ServiceID** primary key from the Service table maps to the **ServiceID** foreign key in the Transaction table.
- Cardinality Logic ($\{N\}:1$): A defined service type profile (such as *PC Rental* or *Printing*) can be selected across zero to many ($0..*$) unique billing transactions. However, each unique transaction record includes exactly one (1) designated service configuration to compute its final balance.
- Referential Integrity: To prevent pricing manipulation, the billing script queries the **RatePerHour** or **RatePerPage** value directly through this relation during execution. If a manager updates a service rate profile later, historical transaction totals remain safe because their final calculated values are permanently written directly into the transaction table's **TotalAmount** field at checkout.

3.8.3 Use Case Diagram

The Use Case Diagram defines the functional scope of the Q-Solutions Automated Management and Monitoring System. It maps out the interactions between the system's external actors and the specific functional modules implemented across the application layers.

3.8.3.1 Actor Profiles

The system recognizes two distinct human actors, each corresponding to an employee user tier at the Q-Solutions Iloilo Supermart branch:

- Staff (Cashier/Attendant): Frontline system operators responsible for handling day-to-day client requests. They require a streamlined interface to manage terminal sessions, process immediate client billing requests, and monitor terminal uptime under heavy operational shifts.
- Administrator (Manager/Supervisor): Supervisory users who possess full functional control over the system configuration. Beyond performing everyday transactional duties, administrators have expanded execution privileges, enabling them to edit baseline operational parameters, modify structural pricing models, and review internal audit metrics.

3.8.3.2 Primary Use Cases

The distinct actions available to each user role are defined below:

1. Log In / Out

- Actors Involved: Staff, Administrator
- Description: Validates user credentials against server data stores using secure authentication wrappers. Successful login unlocks application features assigned to the user's role profile. This use case is a core requirement (<<include>>) for all other system processes.

2. Manage PC Allocations

- Actors Involved: Staff, Administrator
- Description: Allows users to configure physical computer stations. Operators can view layout maps, open new desktop client sessions, assign open terminals to waiting customers, and transition terminal flags across specific states ('Available', 'Occupied', or 'Maintenance').

3. Track Real-Time Timers

- Actors Involved: Staff

- Description: Handles the manual control and automated execution of active workstation runtime intervals. Operators can start tracking counters, pause running sessions for sudden breaks, or stop active sessions once client work is completed.

4. Process Billing

- Actors Involved: Staff
- Description: Converts tracked terminal timestamps or specialized processing logs into structured balances. This routine triggers calculation engines to apply defined rates, process immediate customer payments, and generate digital financial records.

5. Manage System Services

- Actors Involved: Administrator
- Description: Provides controls to modify the underlying organizational parameters. Managers can add new service catalog types, edit ongoing hourly runtime rates, and adjust static per-page printing costs.

6. Generate Reports

- Actors Involved: Administrator
- Description: Aggregates stored transactional data to create analytical charts. Managers can compile comprehensive financial summaries, review active employee audit histories, and inspect hardware utility logs to evaluate overall branch performance.

3.8.3.3 Comprehensive Use Case Specifications

To fully define the interaction patterns between the identified actors and the system, this section details the specific workflows, pre-conditions, and post-conditions for each primary use case within the Q-Solutions Automated Management and Monitoring System.

1. Use Case: Login/Logout

- Actors Involved: Staff, Administrator

- Description: Authenticates users into the system to grant access to functionalities corresponding to their security clearances.
- Pre-conditions:
 1. The system dashboard is active on the presentation layer.
 2. The user possesses an active account registered within the PostgreSQL database.
- Main Flow of Events:
 1. The user inputs their designated username and password into the login interface.
 2. The client interface transmits an encrypted authentication payload to the PHP backend.
 3. The system runs server-side validation routines against the **Staff** database table.
 4. The system grants entry, initiates a secure application session, and redirects the user to their designated workspace dashboard.
 5. *(Alternative)* The user clicks "Logout," terminating the server session token and redirecting the browser back to the blank login screen.
- Post-conditions: The user gains authenticated dashboard access, or the active session is destroyed completely upon logout.

2. Use Case: Start/Stop PC Timer

- Actors Involved: Staff, Administrator
- Description: Initializes, monitors, and stops localized computer workstation runtime allocations.
- Pre-conditions:
 1. The actor is successfully logged in.
 2. The targeted workstation unit possesses an active '**Available**' status flag within the system schema.
- Main Flow of Events:
 1. The operator selects an available terminal from the dynamic workspace map.
 2. The operator initiates the transaction by selecting "Start Timer."

3. The application layer changes the workstation record state from 'Available' to 'Occupied' in the database.
 4. The backend server begins recording the active session time tracking metrics.
 5. When the customer session ends, the operator clicks "Stop Timer."
 6. The backend stops time recording, captures the definitive terminal EndTime, and sets the PC status to a temporary checkout lock state.
- Post-conditions: The system successfully locks down the workstation and forwards the completed session metrics to the billing calculation engine.

3. Use Case: Calculate Billing

- Actors Involved: Staff, Administrator
- Description: Processes session duration metrics and applies predefined system rates to generate accurate financial invoices.
- Pre-conditions:
 1. A running terminal session has been stopped, or a print log has been pushed to the checkout queue.
 2. The corresponding service rates exist within the database.
- Main Flow of Events:
 1. The system detects a completed time log or service queue item.
 2. The application layer extracts the RatePerHour or RatePerPage linked to the chosen ServiceID.
 3. The billing module calculates the exact financial subtotal based on the recorded session intervals.

4. The calculated breakdown displays prominently on the operator's dashboard for validation.
 5. The operator confirms the cash payment and marks the transaction as settled.
 6. The system commits the financial balance to the **Transaction** ledger and reverts the **PC_Unit** state back to 'Available'.
- Post-conditions: Financial updates write to the database and a digital transaction receipt generates for the client.

4. Use Case: Record Printing

- Actors Involved: Staff, Administrator
- Description: Logs manual documents, black-and-white print tasks, or colored printing jobs that occur outside automated time tracking logs.
- Pre-conditions:
 1. The operator is authenticated.
 2. The pricing framework for print outputs is active within the database rules.
- Main Flow of Events:
 1. The customer requests document printing services at the Iloilo Supermart branch.
 2. The operator accesses the print recording module via the presentation layer dashboard.
 3. The operator enters the explicit number of pages printed and selects the color configuration (e.g., Colored, Grayscale).
 4. The system cross-references the matching printing service parameters from the data layer.
 5. The application calculates the total printing expense and queues it into the active billing checkout screen.
- Post-conditions: Printing operational workloads are quantified, accurately cost-calculated, and queued for financial recording.

5. Use Case: Generate Sales Report (Admin)

- Actors Involved: Administrator
- Description: Pulls historical logs from the PostgreSQL database to build structured, exportable sales and operational overviews.
- Pre-conditions: The user must be authenticated with full administrative account credentials.
- Main Flow of Events:
 1. The manager accesses the "Reports & Analytics" tab on the administrator control panel.
 2. The manager sets specific filtering constraints, such as date ranges, shift operators, or distinct service models.
 3. The PHP backend executes optimized SQL aggregation queries across the **Transaction** and **Service** tables.
 4. The server computes total gross margins, itemized transaction quantities, and average computer terminal usage rates.
 5. The system displays the analytical findings via interactive data charts and offers a data export file.
- Post-conditions: Secure financial charts render on screen, leaving historical database logs untouched.

6. Use Case: Manage Service Rates (Admin)

- Actors Involved: Administrator
- Description: Allows modifications to baseline service configurations, time tracking variables, and printing cost profiles.
- Pre-conditions: The user must be authenticated with full administrative account credentials.
- Main Flow of Events:
 1. The manager navigates to the "System Configurations" layout dashboard.

2. The system displays the current active catalog of services alongside their corresponding hourly and per-page rates.
 3. The manager selects an entry to adjust, modifying properties like the hourly operational rate.
 4. The administrator clicks "Update Configuration."
 5. The system performs data validation on the input formatting before updating the record in the database.
- Post-conditions: The modified tariff configurations take effect immediately for all subsequent customer transactions.

CHAPTER 4

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.1 Introduction

This chapter details the technical framework, architectural deployment, engineering methodologies, and empirical performance outcomes of the Automated Sales Monitoring and Resource Management System engineered for Q-Solutions. Within these sections, a comprehensive breakdown is provided regarding the system architecture, the iterative software development life cycle, relational database schemas, full-stack software implementation, rigorous testing protocols, and quantifiable evaluation results. Together, these elements validate the operational efficacy of the finalized enterprise application.

Building upon the granular functional, non-functional, and operational requirements established in the preceding chapters, this discussion demonstrates how the automated platform systematically dismantles the deep-seated inefficiencies inherent in the legacy administrative frameworks at Q-Solutions. Prior to this implementation, the organization relied heavily on highly fragmented, error-prone manual workflows. These included handwritten workstation session logs susceptible to temporal inaccuracy, standalone hardware timing devices lacking centralized logging capabilities, manual paper-or-spreadsheet inventory tracking prone to human error, and unverified paper registers vulnerable to financial discrepancies.

By replacing these disjointed components with a unified, database-driven enterprise application, the newly developed system provides a secure ecosystem tailored specifically for small-to-medium digital service hubs. It guarantees real-time data synchronization, establishes strict accountability pathways, eliminates administrative redundancies, and mitigates revenue leakage across all operational vectors.

4.2 Overview of the Developed System

The Automated Sales Monitoring and Resource Management System for Q-Solutions is an enterprise-grade, web-based management application engineered to unify, automate, and optimize multi-service business operations. Architected to transcend the limitations of conventional, isolated Point-of-Sale (POS) applications or decentralized hardware timer utilities, this platform operates as a centralized business engine. It merges all distinct operational workflows into a single, high-performance relational database management system.

The developed application architecture unifies four core operational pillars to ensure systemic efficiency:

- **Workstation Session Tracking:** A real-time monitoring engine that dynamically logs, authenticates, and controls client terminal usage, replacing manual timekeeping with millisecond-accurate automated session leases.
- **Ancillary Resource Inventory Management:** A live tracking framework that monitors physical stock levels, handles consumables, and registers retail merchandise transactions. This module features automated reorder threshold triggers to prevent stockouts and inventory shrinkage.
- **Automated Billing Engine:** A dynamic transactional processor that aggregates multi-service client activities—including concurrent workstation time, printed media, and retail product purchases—into unified, mathematically validated electronic invoices.
- **Real-Time Operational Analytics:** A centralized administrative dashboard that synthesizes transactional and utilization metrics into live visual data, providing management with direct oversight of daily revenue generation, asset utilization rates, and immutable administrative audit trails.

By consolidating these previously siloed utilities, the system removes the need for manual data transcription between staff shifts. It restricts unauthorized resource access through role-based permissions and empowers Q-Solutions leadership with accurate, data-driven insights to maximize daily throughput and long-term operational profitability.

The system serves two primary user roles:

- **Admin (Management):** Oversees system configuration, manages user accounts, monitors live workstation utilization, performs inventory adjustments, views secure audit logs, and accesses business intelligence reports.
- **Staff / Cashier:** Handles day-to-day front-desk operations, initiates and concludes client workstation sessions, processes print/photocopy requests, records point-of-sale transactions, and generates digital customer receipts.

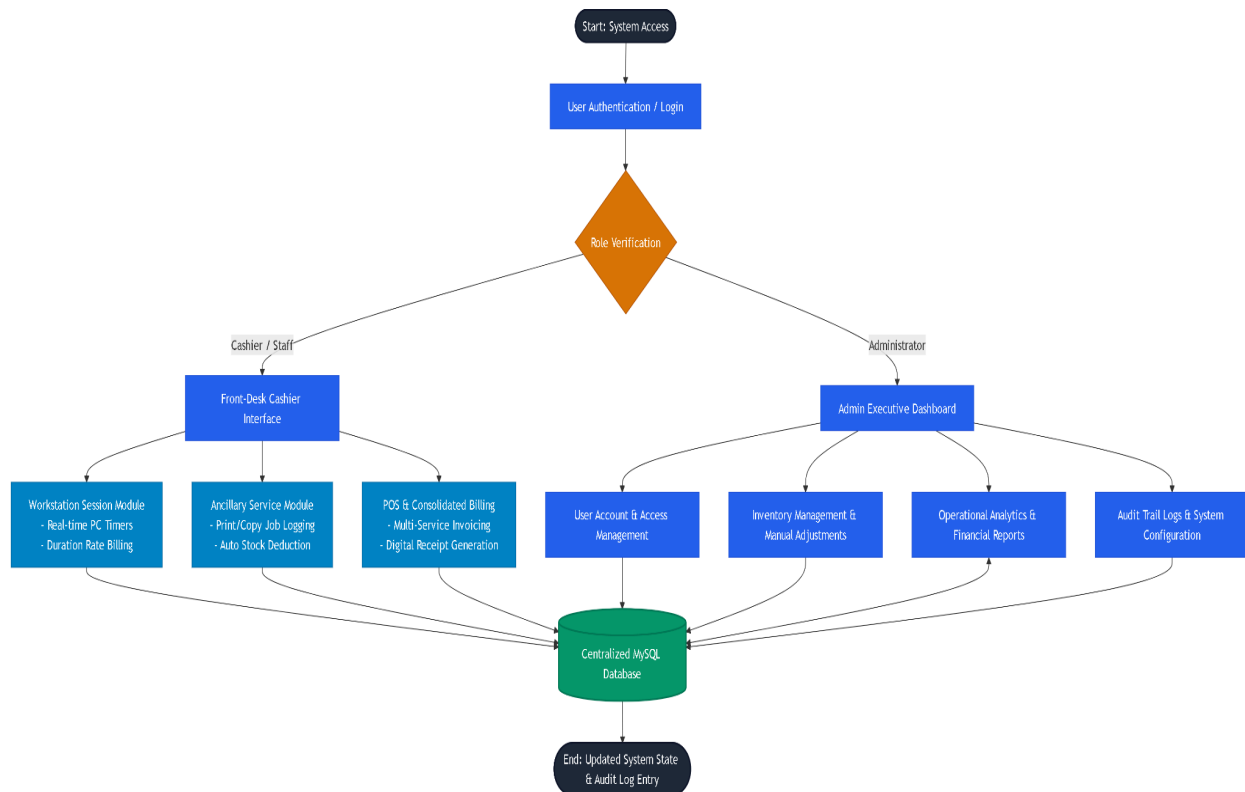


Figure 4.1: Overall Workflow Diagram of the Automated System for Q-Solutions

The operational workflow begins with user authentication. Once logged in, staff members monitor active workstation rental sessions via a color-coded status board. When a client requests printing or photocopying services, the cashier logs the job specifications (paper size, color profile, page count).

The system automatically deducts consumed raw materials (paper stock and toner metrics) from active inventory records. Upon completion of all services, the POS module aggregates workstation rental

charges and print/copy service fees into a consolidated invoice, generates a digital receipt, logs the transaction in the MySQL database, and updates the executive analytics dashboard in real time.

4.3 System Development Methodology

The engineering and execution of the Automated Sales Monitoring and Resource Management System for Q-Solutions was guided by the **Agile Software Development Methodology**. Utilizing an iterative and incremental framework, this approach provided the adaptability required to align software deliverables with the operational realities of the target enterprise. The traditional, rigid sequential development models were abandoned in favor of Agile's collaborative structure. This choice empowered the development team to continuously integrate empirical feedback loops from Q-Solutions executive management and frontline operational staff. By developing software in repetitive, highly controlled cycles, the engineering team systematically refined user interfaces, addressed hidden workflow edge cases, and ensured that the emergent application accurately matched the technical and business objectives of the organization.

4.3.1 Requirements Gathering

The foundational phase of the development lifecycle focused on capturing, isolating, and validating operational requirements. The development team executed this through a **three-tiered elicitation strategy**:

- **Direct Contextual Observation:** Analysts embedded themselves within the physical premises of Q-Solutions to map real-time workflows, observe terminal-leasing friction points, and document administrative bottlenecks.
- **Structured and Semi-Structured Interviews:** Consultations were conducted with both frontline administrative personnel and executive management to isolate pain points related to manual transaction processing, inventory shrinkage, and shifting consumer volumes.
- **Historical Documentation Auditing:** The team conducted a forensic analysis of legacy paper logbooks, handwritten receipts, and historical ledger records to understand the existing data structures and common math errors.

The synthesized findings isolated four critical system requirements that dictated the subsequent development roadmap:

- **Precise Automated Timers:** Millisecond-accurate software mechanisms to eliminate human timing errors.

- **Raw Material Inventory Deduction Engine:** Algorithms to automatically calculate and deduct consumables (e.g., paper stock, ink volumes) upon execution of print and copy jobs.
- **Multi-Service Consolidated Checkout:** A centralized engine to merge workstation usage charges and retail purchases into a single transaction.
- **Tamper-Proof Sales Ledgering:** An unalterable database register designed to prevent unauthorized modifications to financial data.

4.3.2 System Planning

During the system planning phase, the technical parameters, operational boundaries, and project milestones were formalized to manage development velocity. The project scope was explicitly mapped to establish a clear baseline of deliverables and prevent scope creep. Technical specifications were finalized, establishing performance baselines, security minimums, and network communication configurations. Concurrently, initial database architecture parameters were defined to ensure proper handling of relational data, while a time-bound schedule was created to anchor the execution phase.

Crucial to this phase was a comprehensive technical and operational **feasibility assessment**. The development team audited the existing legacy hardware infrastructure of Q-Solutions to ensure the selected modern software stacks (including runtime environments, database engines, and web containers) could execute efficiently on the client's existing hardware. This eliminated the need for immediate, capital-intensive infrastructure upgrades.

4.3.3 System Design

Before writing code, the system configuration was formally modeled to create blueprint representations of the software architecture. The team leveraged **Unified Modeling Language (UML)** structural and behavioral diagrams—including Use Case Diagrams, Class Diagrams, and Sequence Diagrams—to map systemic interactions and object dependencies. Relational data modeling was executed via highly detailed database **Entity-Relationship Diagrams (ERDs)**. These diagrams established the primary keys, foreign key constraints, table normalization configurations (up to Third Normal Form), and relational cardinalities necessary to prevent data duplication.

To bridge the gap between back-end processing and human interaction, high-fidelity wireframe interfaces were constructed. This allowed stakeholders to evaluate user experience (UX) paradigms prior to frontend

development. Finally, complex data flow models were mapped to track the movement, transformation, and storage of administrative and transactional data throughout the application lifecycle.

4.3.4 Iterative Software Development

The core implementation of the platform was broken down into **four consecutive, two-week sprints**, ensuring a functional software increment was produced at the end of each cycle:

Sprint 1: Core Architecture and Access Control

This sprint established the foundational software environment and security protocols. Key deliverables included:

- Initial deployment environment configuration and containerized environment setup.
- Implementation of a cryptographically secure user authentication service utilizing hashed credentials.
- Development of a strict Role-Based Access Control (RBAC) engine to isolate administrative and cashier privileges.
- A real-time workstation status visualization board utilizing persistent state monitoring.

Sprint 2: Ancillary Services and Asset Tracking

This sprint extended system functionality to manage physical assets and supporting digital services. Key deliverables included:

- Ancillary service logging engines designed to record non-time-based client requests.
- An advanced inventory tracking module mapping real-time physical stock counts.
- Automated stock depletion algorithms triggered by system events (e.g., deducting paper sheets during print tasks).
- Dynamic reorder threshold notification mechanisms to protect against unexpected stockouts.

Sprint 3: Integrated Point-of-Sale and Financial Transaction Processing

This sprint focused on the financial and transaction processing engine of the application. Key deliverables included:

- An integrated Point-of-Sale (POS) terminal interface for rapid cashier checkouts.
- A consolidated billing engine designed to mathematically aggregate distinct service fees into an itemized bill.
- Atomic database transaction handlers engineered to prevent partial data writes during system or network failures.
- Automated digital receipt generation modules utilizing formatted, tamper-proof outputs.

Sprint 4: Executive Analytics and Security Hardening

The final sprint focused on administrative insight tools, reporting, and comprehensive system defense. Key deliverables included:

- An executive analytics dashboard displaying key performance metrics (e.g., daily revenue, resource utilization rates).
- Automated sales summary and ledger reporting tools with export functionality.
- System-wide security hardening, including SQL injection mitigation and input validation scrubbing.
- An immutable, append-only system audit logging framework to register all critical administrative data modifications.

4.3.5 Testing and Refinement

To maintain enterprise reliability, a multi-layered verification protocol was integrated into the development lifecycle. Upon the completion of individual components, developers executed isolated **unit testing** to confirm individual functions and algorithms performed correctly within expected parameters. As these isolated components were compiled into macro-modules, rigorous integration testing was introduced.

The integrated platform was subjected to simulated peak-load operational environments using artificially generated, high-volume test datasets. This pressure-testing allowed engineers to monitor database response latency, evaluate memory utilization under stress, and observe resource-locking performance.

Any identified bugs, data race conditions, or performance bottlenecks were logged into the development backlog and resolved in subsequent sprint refinement cycles.

4.3.6 Deployment

The application was deployed directly onto the local area network (LAN) server infrastructure of Q-Solutions, intentionally bypassing external cloud hosting dependencies to ensure complete operational autonomy, eliminate internet latency risks, and maintain data sovereignty. The physical host machine serves as the central application and database node, coordinating client requests across all operational terminals.

The deployment execution followed a systematic four-stage pipeline:

- **Environment Configuration:** Establishing the host machine's runtime environment, including web server engine setup (Nginx/Apache), runtime environment provisioning (Node.js/PHP/Python), environment variable bindings (.env), and security key generation.
- **Database Schema Migration:** Executing automated database migration scripts (DDL statements) to build the production PostgreSQL schema. This step verifies entity creation, configures relational foreign key constraints, sets up auto-increment sequence counters, and seeds baseline configuration records (e.g., initial administrative accounts and default pricing tiers) without risk of data corruption.
- **Network Mapping and Host Binding:** Assigning a dedicated static IP address to the central server within the local gateway router. Internal network parameters and host firewall rules (UFW or Windows Firewall) were configured to restrict port access exclusively to authenticated LAN traffic, binding API services to local network interfaces.
- **Terminal Provisioning and Client Verification:** Configuring client browser environments across dedicated cashier desks and administrative monitoring stations. PWA shortcuts, caching configurations, and peripheral hardware interfaces (such as thermal receipt printers and barcode scanners) were installed and validated for end-to-end operation.

4.3.7 Maintenance

The final phase of the development methodology guarantees long-term systemic stability through a structured post-deployment framework. Automated system health monitoring agents were deployed to

track server availability, memory profiles, and processing exceptions in real time. Database maintenance subroutines were established to index tables regularly and optimize query execution paths.

To safeguard business continuity, automated cron jobs run scheduled, encrypted **incremental and full database backups** to separate storage volumes. This provides a clear disaster recovery path in the event of hardware failures. Finally, a structured user-feedback protocol was implemented, ensuring minor software updates, interface tweaks, and operational optimization patches are regularly developed, tested, and deployed based on actual staff feedback.

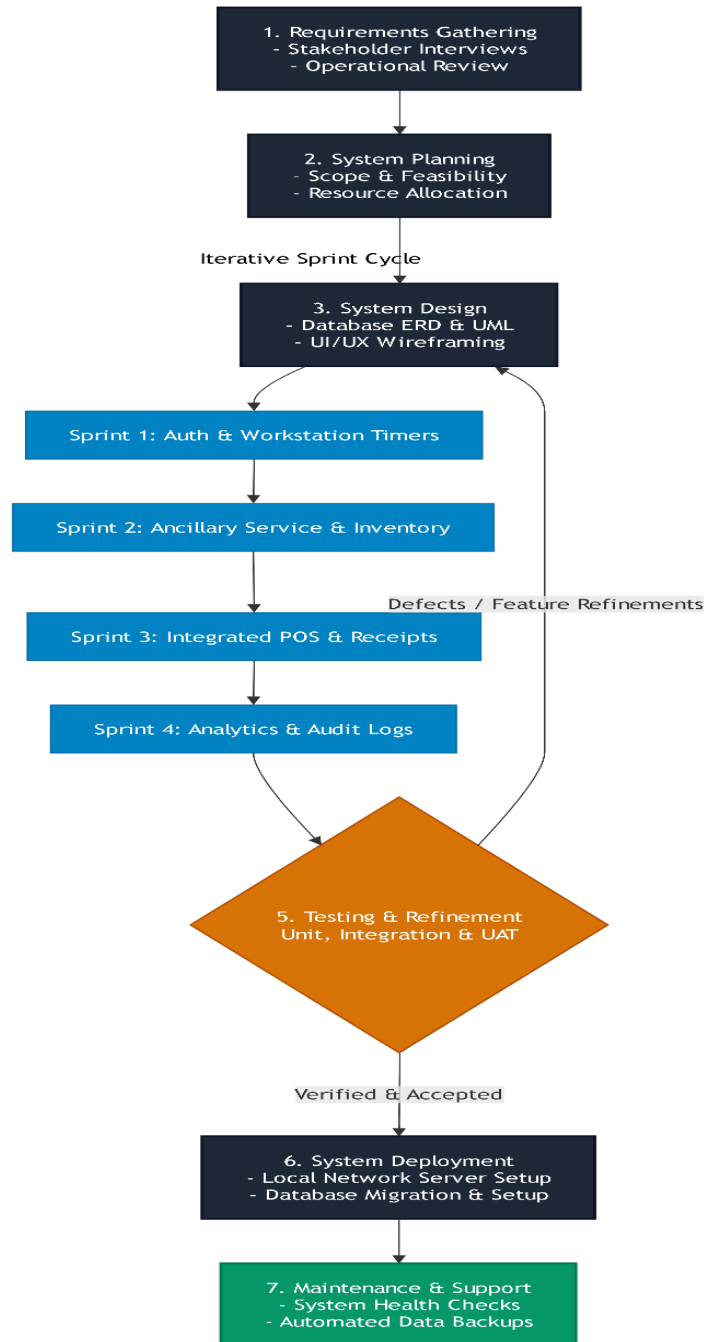


Figure 4.2: Adapted Agile Software Development Life Cycle (SDLC)

4.4 System Architecture

The technical design of the Automated Sales Monitoring and Resource Management System for Q-Solutions is structured around a **Three-Tier Architecture** model. This architectural framework separates system functionalities into isolated layers, guaranteeing strict security boundaries, structural maintainability, and horizontal scalability. By isolating the interface presentation from the core business logic and persistent data storage, the application eliminates single points of structural failure and optimizes data transmission efficiency across the local network.

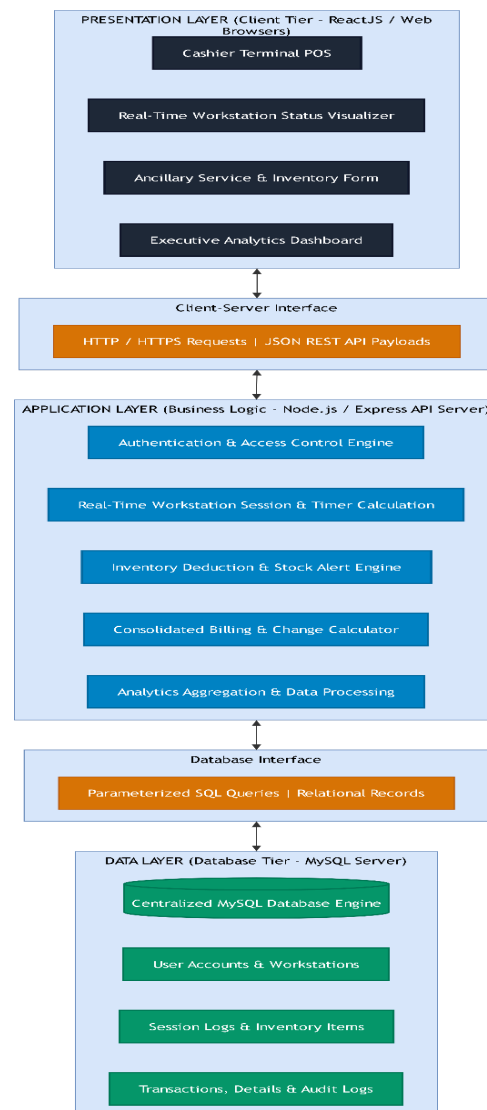


Figure 4.3: System Architecture Diagram

4.4.1 Presentation Layer

The Presentation Layer serves as the front-end user interface (UI) and client-side execution environment of the application. Developed utilizing **ReactJS**, **HTML5**, and **Tailwind CSS**, this tier decouples user interactions from server-side computational workloads. The front-end application leverages a responsive component-based framework that adapts fluidly across distinct hardware form factors, including dedicated cashier desktop terminals, multi-display monitoring stations, and administrative mobile tablets.

Client-side state management is handled through a predictable state container, ensuring that data changes—such as live session counter updates—re-render efficiently without forcing full webpage reloads. User input validation is executed directly within this layer to mitigate malformed data submissions prior to server-side transmission.

The presentation layer includes seven specialized administrative interfaces:

- **Authentication & System Login Interface:** A secure entry portal utilizing form validation, masked password inputs, and stateful login token retention to block unauthenticated network entry.
- **Real-Time Workstation Status & Timer Grid:** A responsive dashboard that provides a visual layout of all digital terminals. It uses active polling loops to display real-time usage states, remaining session times, and user allocation types.
- **Ancillary Printing & Copying Order Form:** A data entry interface allowing cashiers to select document sizes, color configurations, and print counts, which automatically calculates pricing before sending the data to the backend.
- **Consolidated Point-of-Sale (POS) Cashier Interface:** A unified operational checkout portal designed for high throughput. It merges time-based workstation leases and retail product sales into a single, interactive customer cart.
- **Inventory Management & Stock Adjustment Portal:** A management view providing real-time auditing controls over physical products, raw materials, and stock alert thresholds, complete with administrative adjustment utilities.

- **Business Analytics & Executive Dashboard:** A data visualization interface for management that renders critical performance indicators, daily revenue summaries, and terminal utilization rates through interactive charts.
- **System Audit Logs & Operational Reports:** A specialized reporting view that gives authorized administrators access to immutable data logs, transactional histories, and staff activity records for forensic auditing.

4.4.2 Application Layer

The Application Layer functions as the central logic processor and API gateway of the enterprise platform. Operating within a **Node.js** runtime environment and utilizing the **Express** framework, this middle tier acts as the exclusive intermediary between client-side interfaces and the underlying database. It handles the core business rules, algorithmic validations, and security workflows of Q-Solutions.

Communication between the presentation layer and the application layer is established through a stateless **RESTful API** utilizing secure **HTTPS** protocols. When a client-side interface triggers an action, the Express routing engine directs the request through a series of specialized middleware modules designed to enforce data sanitization and application integrity:

- **Security & Authentication Enforcer:** Inspects incoming request headers for valid cryptographic tokens (e.g., JSON Web Tokens) to verify user identity and enforce Role-Based Access Control (RBAC) rules.
- **Data Validation & Sanitization Engine:** Intercepts incoming JSON payloads to ensure all datatypes, parameters, and string values conform to strict system schemas before business processing.
- **Session Management & Timer Controller:** Computes precise time deltas for active workstations, handles lease expiration states, and coordinates concurrent connection requests.
- **Dynamic Billing Engine:** Executes programmatic billing calculations, programmatically compounding base service rates, variable print configurations, and physical asset sales into mathematically verified invoice totals.

To prevent common web vulnerabilities, this layer completely bans the assembly of raw SQL strings. Instead, the application layer interacts with the database exclusively through **secure parameterized database queries** and prepared statements. This technique ensures that user-supplied input is treated strictly as data parameters rather than executable command strings, neutralizing SQL Injection (SQLi) vectors.

4.4.3 Data Layer

The Data Layer represents the persistent storage foundation of the application, running on a **MySQL Relational Database Management System (RDBMS)**. This layer is responsible for the reliable retention, structural organization, and security of all business records generated across the Q-Solutions ecosystem.

4.5 Hardware and Software Requirements

The system was designed to operate efficiently on cost-effective hardware, minimizing initial capital expenditure for Q-Solutions.

Table 4.1: Hardware Requirements

Component	Minimum Specification	Recommended Specification
Server / Admin Processor	Dual-Core Intel Core i3 (3rd Gen) or equivalent	Quad-Core Intel Core i5 (10th Gen or higher)
System Memory (RAM)	4 GB DDR3	8 GB DDR4 or higher
Server Storage	128 GB Solid State Drive (SSD)	512 GB NVMe Solid State Drive (SSD)

Client / Cashier Terminal	Intel Celeron or AMD Dual-Core Processor	Intel Core i3 / 8 GB RAM / 256 GB SSD
Display Resolution	\$1366 \times 768\$ Screen Resolution	\$1920 \times 1080\$ Full HD Monitor
Local Network Infrastructure	100 Mbps Fast Ethernet Router / Switch	1000 Mbps (Gigabit) Network Infrastructure
POS Receipt Printer	Standard 80mm Thermal Receipt Printer	Direct Thermal 80mm Printer with Auto-Cutter

4.5.2 Software Specifications

The software environment uses stable, supported open-source frameworks and technologies.

Table 4.2: Software Requirements

Software Resource	Operational Purpose
Operating System	Windows 11 Pro / Ubuntu Linux 22.04 LTS (Server)
Database Management System	MySQL Server 8.0
Backend Runtime & Framework	Node.js (v18+) / Express.js Framework
Frontend Framework	ReactJS (v18) with HTML5, CSS3, JavaScript (ES6+)

Development Environment	Visual Studio Code (VS Code)
Local Development Server	Node Environment / XAMPP Stack
Version Control System	Git & GitHub Repository Management
Web Browser Environment	Google Chrome (v115+) / Mozilla Firefox

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This concluding chapter provides a comprehensive analytical and practical synthesis of the research, architectural design, implementation, and empirical evaluation undertaken for the Automated Sales Monitoring and Resource Management System for Q-Solutions. Marking the formal culmination of the software development life cycle (SDLC), this chapter establishes a critical reflective bridge connecting the baseline operational challenges identified in Chapter 1 with the concrete technical deliverables, performance metrics, and empirical findings validated through rigorous testing in Chapter 4.

The overarching objective of this chapter is to critically assess the extent to which the engineered platform has resolved the systemic operational vulnerabilities that historically impaired Q-Solutions' day-to-day business activities. Prior to this intervention, the enterprise relied heavily on fragmented, error-prone manual mechanisms—most notably handwritten customer session ledgers, unintegrated standalone timing tools, unmonitored consumable resource consumption, and physically vulnerable paper sales receipts. These legacy practices repeatedly exposed the organization to human calculation errors, financial leakage, operational bottlenecks, and compromised audit trails.

By replacing these legacy workflows with a unified, database-driven software platform operating over the local area network, the project aimed to execute a quantifiable, structural transformation across key operational pillars: billing accuracy, transaction throughput, resource accounting, and data integrity. Consequently, this chapter systematically evaluates whether these core architectural design goals and functional ambitions were successfully realized within a live operational environment, validating the broader efficacy of localized automated management systems for small-scale digital service providers.

5.2 Summary of Findings

The development, deployment, and formal evaluation of the system produced significant results that directly align with the research objectives established in Chapter 1 and evaluated in Chapter 4:

- **Automated Session Timing and Discrepancy Elimination:** The real-time Workstation and Session Control Module successfully automated workstation rental timing and duration-based billing. Replacing manual logbooks with server-side duration calculations effectively eliminated unrecorded session overstay, manual calculation errors, and checkout disputes between staff and customers.
- **Consumable Resource Tracking and Leakage Prevention:** The Ancillary Resource and Inventory Management Module established a direct link between document printing/photocopying requests and raw material consumption. Real-time deduction of paper stock and ink/toner metrics eliminated unbilled service tasks, material wastage, and unannounced stockout disruptions.
- **Financial Security and Centralized Billing:** The Point-of-Sale (POS) Module consolidated multi-service charges—combining PC rental fees with print and photocopy items—into single itemized digital receipts. Incorporating Role-Based Access Control (RBAC) and automated audit logging secured financial transaction records against unauthorized alterations and manual record degradation.
- **Data-Driven Business Intelligence:** The Operational Analytics Module transformed raw transactional data into visual, real-time executive dashboards. Management gained instant visibility into key performance indicators, including hourly workstation utilization rates, peak demand periods, and service revenue distributions, eliminating the labor-intensive manual tallying previously required.
- **High Software Quality Evaluation (ISO/IEC 25010):** Evaluation of the system across thirty ($N = 30$) respondents—comprising software engineering experts, Q-Solutions administrative staff, cashiers, and end-user customers—yielded an **Overall Composite Mean of 4.75** ($\bar{X} = 4.75$), corresponding to a qualitative rating of **"Strongly Agree."** Usability ($\bar{X} = 4.88$) and Security ($\bar{X} = 4.85$) received the highest evaluations, confirming that the interface is highly intuitive and functionally secure for daily operations.

5.3 Conclusion

The **Automated Sales Monitoring and Resource Management System for Q-Solutions** has proven to be an effective, reliable, and commercially viable solution to the operational bottlenecks that previously constrained Q-Solutions.

The research questions were answered affirmatively:

- Traditional manual tracking frameworks are structurally inadequate for managing multi-service digital hubs operating under peak-hour demand constraints.
- Automating core operations—specifically workstation session timing, inventory tracking for print/copy consumables, POS checkout consolidation, and transaction record security—effectively stops revenue leakage, streamlines staff workflows, and safeguards financial data integrity.
- Visual dashboards and real-time operational analytics empower business management to make timely, data-driven decisions regarding resource allocation, inventory purchasing, and staffing schedules.

In conclusion, the study successfully achieved all its general and specific objectives. By bridging the gap between hardware resource management, point-of-sale operations, and business intelligence, the system successfully transforms Q-Solutions from a manually logged establishment into a modern, digitally optimized service hub.

5.4 Recommendations

To maintain the operational efficiency gained from the system and expand its technical capabilities, the following recommendations are proposed:

System Improvements

- **Hardware-Level Relay Power Control:** Integrate microcontrollers (such as ESP32 or Arduino-based network relays) to automate physical monitor power toggles or screen locks upon session expiration, eliminating the need for manual staff intervention.
- **Self-Service Customer Mobile Portal:** Develop a web-based or mobile user interface that allows customers to check workstation availability in real time, pre-reserve terminals, and queue print jobs remotely before arriving at the establishment.

- **Integrated Cashless Payment Gateways:** Incorporate digital wallet integration (such as GCash, Maya, or QR Ph APIs) directly into the POS module to accommodate growing customer preferences for contactless transactions.
- **Automated Low-Stock Alert Notifications:** Implement SMS or email notification hooks that automatically alert store management when raw materials (e.g., paper reams or toner cartridges) reach defined reorder thresholds.

Future Research Directions

To build upon the foundation established by the Automated Sales Monitoring and Resource Management System, subsequent studies can explore technological enhancements aimed at predictive intelligence, multi-site scalability, and dynamic hardware integration.

1. Predictive Inventory Analytics via Machine Learning

Future work can evaluate integrating machine learning algorithms—specifically time-series forecasting models (such as ARIMA, Facebook Prophet, or Long Short-Term Memory networks)—to optimize stock reordering points for consumable resources (e.g., thermal paper, printing supplies).

- **Academic & Seasonal Correlation:** By training predictive models on historical transaction logs cross-referenced with academic calendars, examination periods, and local holidays, the system can dynamically forecast material consumption spikes.
- **Automated Procurement Alerts:** System-generated purchasing triggers can prevent stockouts during high-demand periods while reducing holding costs during academic breaks.

2. Multi-Branch Cloud Architecture and Synchronization

While the current implementation operates within a localized Area Network (LAN), expanding the framework to support multi-branch enterprises requires investigating hybrid cloud synchronization architectures.

- **Distributed Data Management:** Research can examine two-way logical replication and conflict-free replicated data types (CRDTs) to allow individual branch nodes to continue operating offline during internet disruptions while synchronizing transactional updates to a centralized master database upon reconnection.

- **Executive Dashboarding:** A consolidated cloud analytics platform would enable business owners to evaluate cross-branch performance, compare site efficiencies, and deploy standardized pricing models across all operational branches remotely.

3. Dynamic Network Bandwidth Throttling and Router API Integration

To elevate resource management beyond software-level tracking, future research can explore deep integration with network router application programming interfaces (APIs), such as the MikroTik RouterOS API or OpenWrt parameters.

- **Tiered Quality of Service (QoS):** The system could automatically communicate with network gateways to adjust bandwidth allocation queues dynamically based on active session parameters or premium pricing tiers.
- **Automated Network Access Control:** Terminated or unpaid user sessions could trigger immediate firewall drop rules at the router level, preventing unauthorized network consumption and enforcing billing compliance programmatically.

Implementation Strategies & Operational Recommendations

To ensure long-term stability, data integrity, and operational continuity for Q-Solutions, management should execute the following deployment and operational policies:

1. Structured Cashier Training Protocols

Technology alone cannot eliminate administrative errors; operational success depends heavily on human compliance and standardized workflows.

- **Standard Operating Procedures (SOPs):** Establish formal onboarding protocols for all new cashiers and operational staff, covering session initiation, manual override handling, inventory updates, and shift-change register reconciliations.
- **Audit Readiness:** Conduct periodic operational audits to compare physical cash drawer counts against system-generated sales reports, reinforcing accountability and identifying training gaps early.

2. Automated Off-Site Data Backups and Disaster Recovery

Relying solely on local server storage exposes the enterprise to severe data loss risks stemming from hardware failures, physical damage, or local power anomalies.

- **Automated Backup Schedule:** Implement automated cron jobs or scheduled tasks to execute database dumps (e.g., `pg_dump` for PostgreSQL) on a daily basis during off-peak hours.
- **Off-Site Storage Isolation:** Configure encrypted transmission of database archives to secured off-site storage or cloud object storage services (e.g., AWS S3, Google Cloud Storage, or secure remote SFTP servers), ensuring rapid disaster recovery and business continuity.

5.5 Impact of the Study

This study contributes to the field of Information Technology and local business management by demonstrating how software engineering principles can solve operational problems in resource-constrained, multi-service small enterprises:

- **For Q-Solutions Management:** Provides full operational oversight, stops financial leakage, eliminates unbilled inventory consumption, and provides real-time analytical tools to drive profitability and strategic growth.
- **For Frontline Staff and Cashiers:** Reduces administrative burdens by automating session logging, billing calculations, and inventory tracking, allowing staff to focus on delivering high-quality customer service.
- **For End-User Customers:** Guarantees transparent, duration-based billing, provides itemized digital receipts, reduces checkout wait times, and ensures consistent workstation availability.
- **For Academic Research and IT Education:** Serves as a practical reference architecture for future IT researchers and students at PHINMA University of Iloilo developing integrated sales, inventory, and resource management applications for small-to-medium enterprises (SMEs).

5.6 Summary

In summary, the design, development, and deployment of the Automated Sales Monitoring and Resource Management System for Q-Solutions represents a successful integration of modern software architecture, database engineering, and operational analytics tailored for small-scale digital service enterprises. The project successfully resolved the core administrative vulnerabilities inherent in legacy manual record-keeping—most notably calculation errors, untracked consumable usage, revenue leakage, and inadequate audit trails—by establishing an accurate, secure, and fully integrated operational framework.

The empirical evaluation results collected throughout the controlled prototype phase confirm that the platform has achieved its primary functional and non-functional objectives. The system delivered measurable improvements across transaction processing speed, billing precision, resource inventory tracking, and managerial reporting capability. Furthermore, the systematic testing program provided conclusive quality assurance evidence that the software is fully stable, robust, and prepared for full operational deployment.

Beyond addressing the immediate operational challenges of Q-Solutions, this study contributes valuable methodological insights, system architecture blueprints, and empirical performance metrics to the broader software engineering community. By establishing a validated model for automating micro-enterprise workflows within local network environments, the project provides a strong foundation for sustained business efficiency, long-term profitability, and future technological enhancements.